

EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 1B

1 KVL/KCL Review

Kirchhoff's Circuit Laws are two important laws used for analyzing circuits. Kirchhoff's Current Law (KCL) says that the sum of all currents entering a node must equal 0. For example, in Figure 1, the sum of all currents entering node 1 is $I_1 - I_2 - I_3 = 0$. Assuming that I_1 and I_3 are known, we can easily obtain a solvable equation for V_x by applying Ohm's law: $I_1 - \frac{V_x}{R_1} - I_3 = 0$.

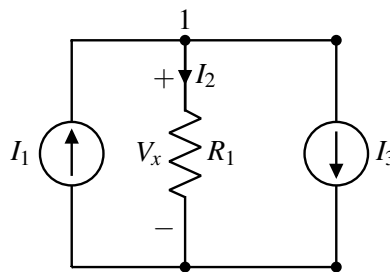


Figure 1: KCL Circuit

Kirchhoff's Voltage Law (KVL) states that the sum of all voltages in a circuit loop must equal 0. To apply KVL to the circuit shown in Figure 2, we can add up voltages in the loop in the counterclockwise direction, which yields $-V_1 + V_x + V_y = 0$. Using the relationships $V_x = i \cdot R_1$ and $i = I_1$, we can solve for all unknowns in this circuit. You can use these two laws to solve any circuit that is planar and linear.

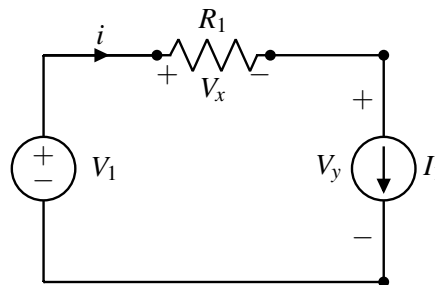


Figure 2: KVL Circuit

If you would like to review these concepts more in-depth, you can check out [the EECS16A Fall 2020 course notes](#).

2 Op-amp Review

Figure 3 shows the equivalent model of an op-amp. It is important to note that this is the general model of an op-amp, so our op-amp golden rules cannot be applied to this unless certain conditions are met.

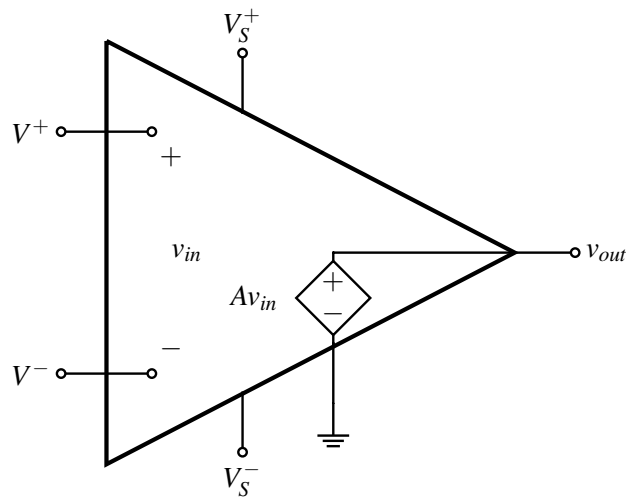


Figure 3: General Op-Amp Model

Conditions Required for the Golden Rules:

- (a) $R_{in} \rightarrow \infty$
- (b) $R_{out} \rightarrow 0$
- (c) $A \rightarrow \infty$
- (d) The op-amp must be operated in negative feedback.

When conditions (a)-(c) are met, the op-amp is considered ideal. Figure 4 shows an ideal op-amp in negative feedback, which can be analyzed using the Golden Rules.

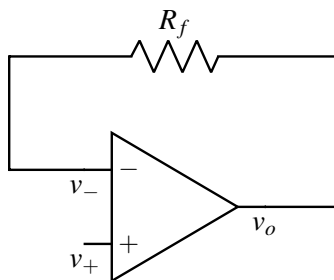


Figure 4: Ideal Op-Amp in Negative Feedback

Golden Rules of ideal op-amps in negative feedback:

- (a) No current can flow into the input terminals ($I_- = 0$ and $I_+ = 0$). (This property follows from condition (a) and does not require negative feedback.)
- (b) The (+) and (-) terminals are at the same voltage ($V_+ = V_-$).

If you would like to review these concepts more in-depth, you can check out [op-amp introduction](#) and [op-amp negative feedback](#) from the EECS16A course notes.

1. KVL/KCL Review

Use Kirchhoff's Laws on the circuit below to find V_x in terms of V_{in}, R_1, R_2, R_3 .

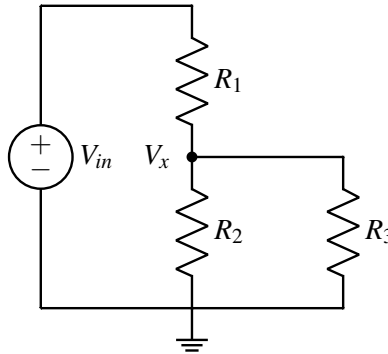


Figure 5: Example Circuit

- What is V_x ?
- As $R_3 \rightarrow \infty$, what is V_x ? What is the name we used for this type of circuit?

2. Op-Amp Summer

Consider the following circuit (assume the op-amp is ideal):

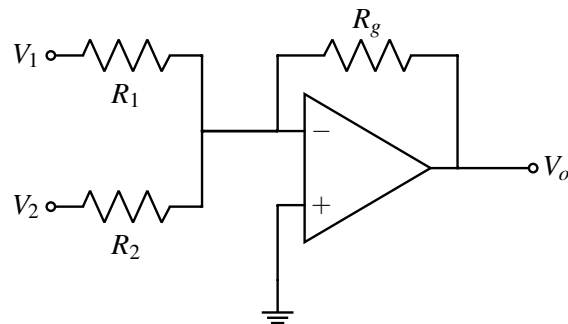


Figure 6: Op-amp Summer

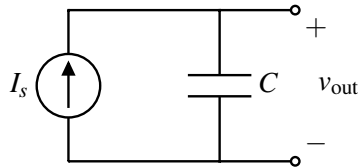
What is the output V_o in terms of V_1 and V_2 ? You may assume that R_1 , R_2 , and R_g are known.

3. Current Sources And Capacitors (The following problem has been adapted from EECS16A Fall 20 Disc 9A.)

Recall charge has units of Coulombs (C), and capacitance is measured in Farads (F) = $\frac{\text{Coulomb}}{\text{Volt}}$.

It may also help to note metric prefix examples: $3\mu\text{F} = 3 \times 10^{-6}\text{F}$.

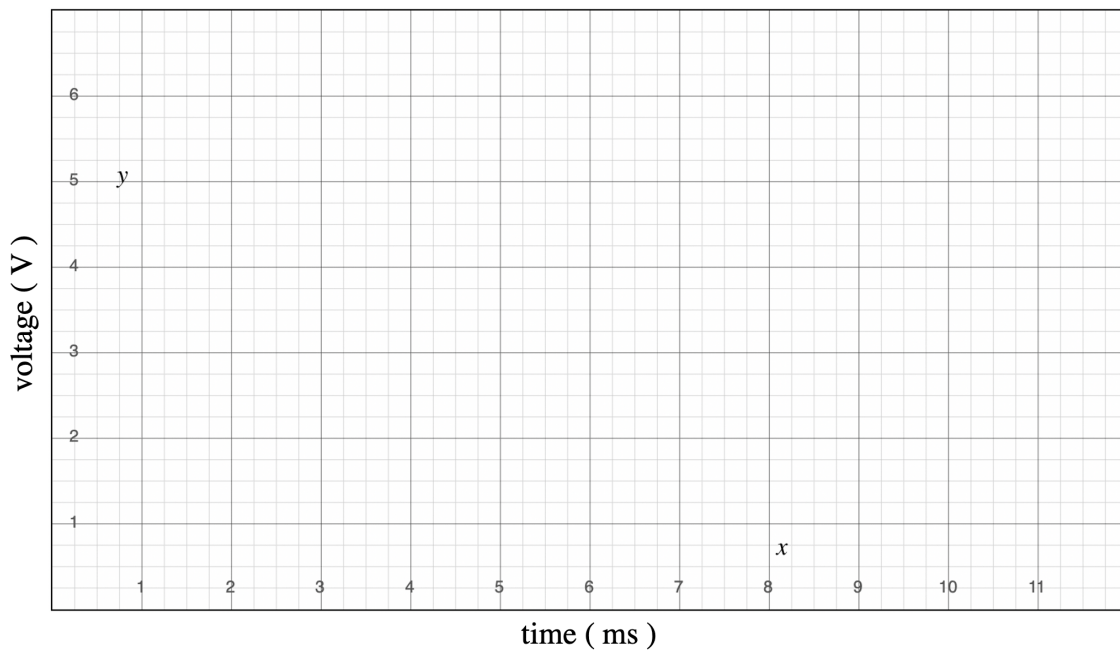
Given the circuit below, find an expression for $v_{\text{out}}(t)$ in terms of I_s , C , V_0 , and t , where V_0 is the initial voltage across the capacitor at $t = 0$.



Then plot the function $v_{\text{out}}(t)$ over time on the graph below for the following conditions detailed below. Use the values $I_s = 1\text{mA}$ and $C = 2\mu\text{F}$.

- Capacitor is initially uncharged $V_0 = 0$ at $t = 0$.
- Capacitor has been charged with $V_0 = +1.5\text{V}$ at $t = 0$.
- Practice:** Swap this capacitor for one with half the capacitance $C = 1\mu\text{F}$, which is initially uncharged $V_0 = 0$ at $t = 0$.

HINT: Recall the calculus identity $\int_a^b f'(x)dx = f(b) - f(a)$, where $f'(x) = \frac{df}{dx}$.



4. Linear Algebra Review

For the following matrices, find the following properties:

- i. What is the column space of the matrix?
- ii. What is the null space of the matrix?
- iii. What are the eigenvalues and corresponding eigenspaces for the matrix?

(a) $\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -2 \\ 2 & -4 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}$

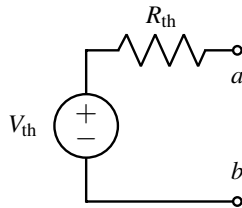
For this matrix you are told that the eigenvalues are: 2, 1, and 0.

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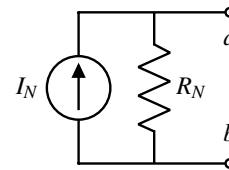
Discussion Worksheet Discussion 2A

1 Circuit Equivalence

To review the circuit equivalence concepts exercised in this worksheet, please see [Note 0B, adapted from the EECS 16A Course Notes](#). We will work through examples in this worksheet. The two forms are presented below for your reference.

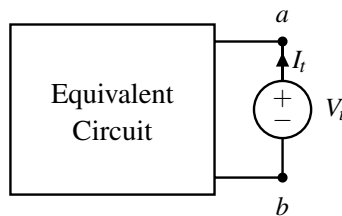


(a) General form of a Thévenin equivalent circuit. Given a circuit and two output terminals, we know the above gives a voltage-source based equivalent; the work lies in solving for V_{th} and R_{th} .

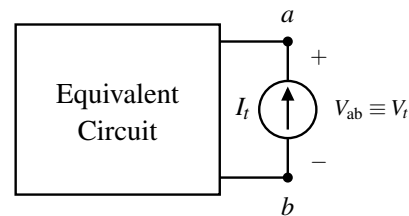


(b) General form of a Norton equivalent circuit, the current-source based equivalent form for a given circuit. Here, we must solve for I_N and R_N .

Below, we display pictorially how to apply current or voltage test sources to a circuit to find R_{eq} .



(a) By applying a test voltage V_t across a and b , we can measure the resulting current draw of the equivalent circuit I_t and use that to calculate R_{eq} .



(b) By feeding a test current I_x into the equivalent circuit, we can measure the resulting voltage drop V_{ab} and calculate R_{eq} .

Figure 2: We can use either a test voltage source or a test current source to find R_{eq} . The choice for what is easier will depend on the specific problem.

2 Transistor Introduction

Transistors (as presented in this course) are 3 terminal, voltage-controlled switches. This means that, when a transistor is "on," the Source (S) and Drain (D) terminals are connected via a low resistance path (short circuit). When a transistor is "off," the Source and Drain terminals are disconnected (open circuit).

Two common types of transistors are NMOS and PMOS transistors. Their states (shorted or open) are determined by comparing the voltage between the G and S terminals ($V_{GS} = V_G - V_S$) to a "threshold voltage"

(V_{tn} for NMOS, V_{tp} for PMOS). Generally, NMOS transistors turn on when V_{GS} is high enough, and PMOS transistors turn on when V_{GS} is low enough (they have complementary behavior!). Transistors are extremely useful in digital logic design since we can use them to implement Boolean logic operators.

In this class, V_{tn} denotes how much **higher** V_G needs to be relative to V_S for the NMOS to be on (allow current flow from drain to source), and $|V_{tp}|$ denotes how much **lower** V_G gate needs to be relative to V_S for the PMOS to be on.

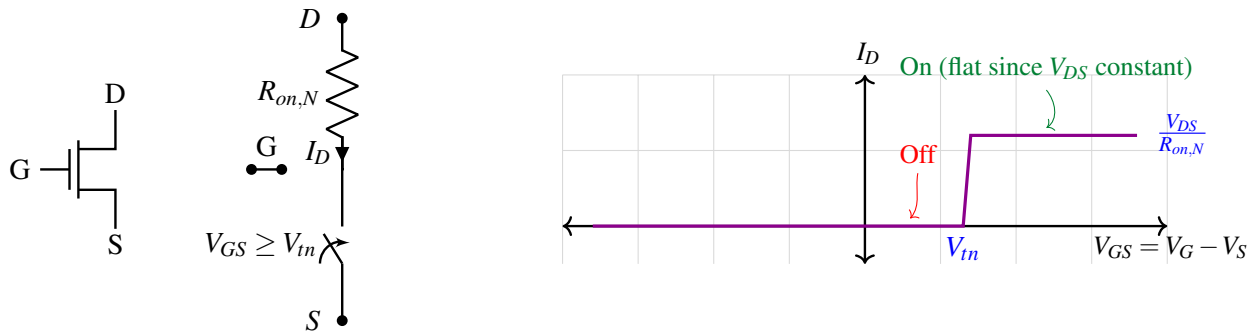


Figure 3: NMOS Transistor Resistor-switch model (the current holding constant at high V_{GS} assumes that V_{DS} is constant).

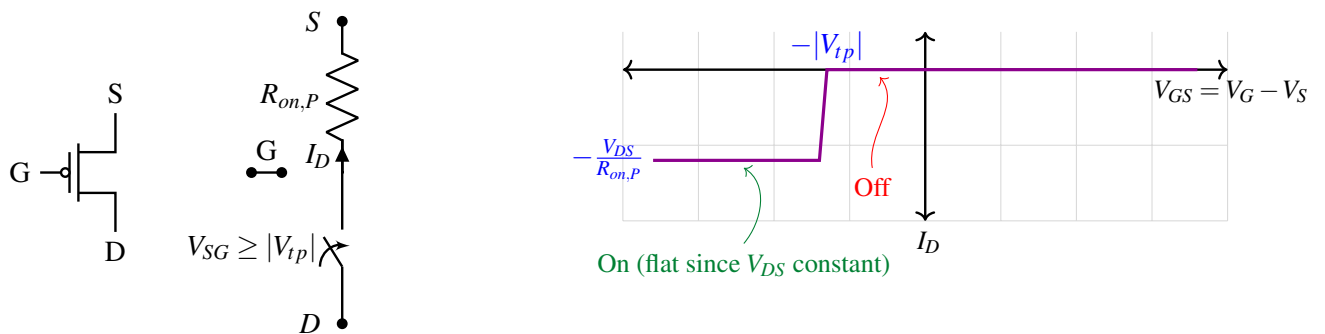


Figure 4: PMOS Transistor Resistor-switch model (the current holding constant at low V_{GS} assumes that V_{DS} is constant). Note that $V_{SG} = -V_{GS}$.

We mentioned that transistors can be connected to perform boolean algebra. An example of this is seen in Section 2, which is called an "inverter" and represents a NOT gate.

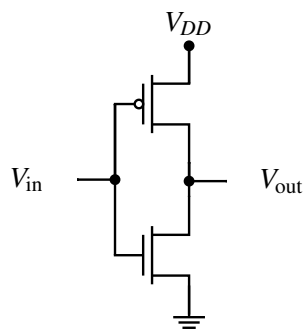


Figure 5: CMOS Inverter

To see why this circuit is called an inverter, we consider the following cases¹:

- (1) **The input is high:** $V_{in} = V_{DD}$. Then, since $V_{GS} \geq V_{tn}$ ($V_{in} \geq V_{tn}$), the NMOS is on. Also, since $V_{GS} \geq V_{tp}$ ($V_{in} - V_{DD} \geq V_{tp} \implies V_{in} \geq V_{DD} - |V_{tp}|$), the PMOS is off. So, only the NMOS switch is closed, and $V_{out} = 0$. That is, **the output is low**.
- (2) **The input is low:** $V_{in} = 0$. Then, since $V_{GS} \leq V_{tn}$ ($V_{in} \leq V_{tn}$), the NMOS is off. Also, since $V_{GS} \leq V_{tp}$ ($V_{in} \leq V_{DD} - |V_{tp}|$), the PMOS is on. So, only the PMOS switch is closed, and $V_{out} = V_{DD}$. That is, **the output is high**.

We can summarize this analysis, using the following truth table:

V_{in}	V_{out}	NMOS	PMOS
V_{DD}	0	on	off
0	V_{DD}	off	on

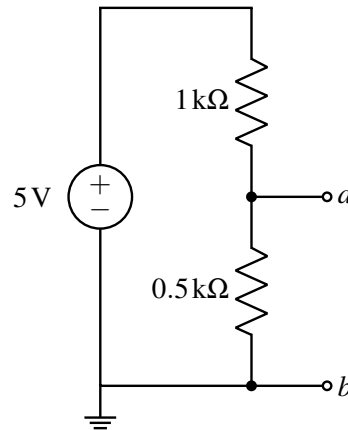
If you think of V_{DD} being a logical 1 and 0V being a logical 0, we have just created the most elementary logical operation using transistors!

¹When working with digital circuits like in Section 2, we usually only consider the values of $V_{in} = 0$ and $V_{in} = V_{DD}$.

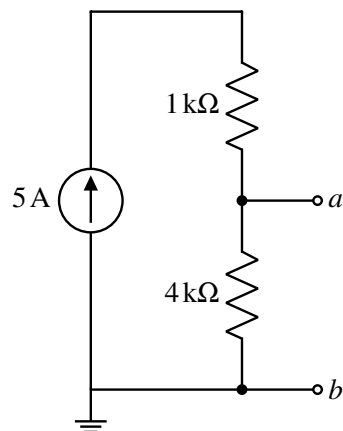
1. Thévenin and Norton Equivalence

Find the Thévenin and Norton equivalents across terminals a and b for the circuits given below. Note that the general forms of these equivalents can be found in Figure 1a and Figure 1b.

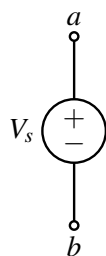
(a)



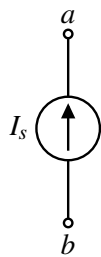
(b)



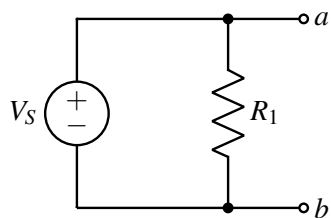
(c) **(Practice)**



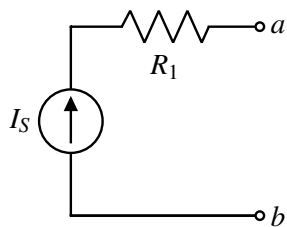
(d) **(Practice)**



(e) **(Practice)**

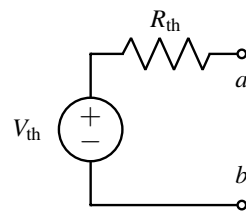
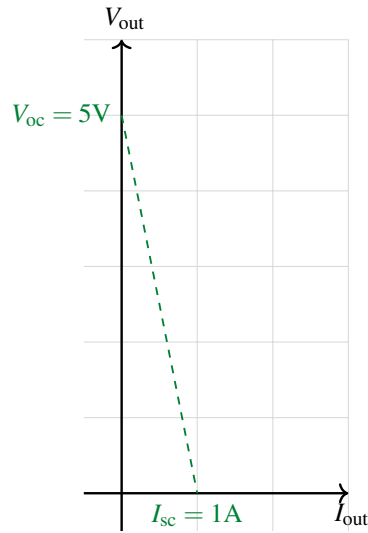


(f) **(Practice)**

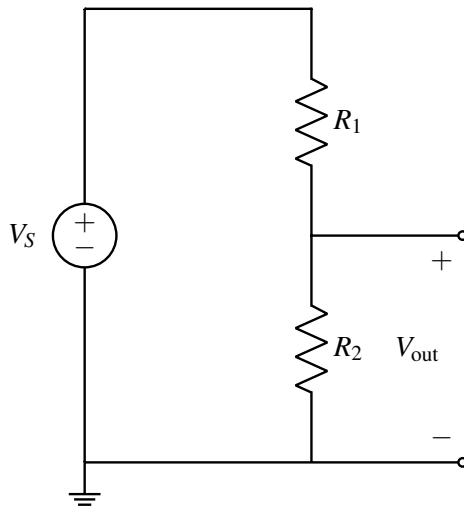


2. Finding Thévenin Equivalents

- (a) You are given the following $I_{\text{out}} - V_{\text{out}}$ characteristic of the Thévenin model of a circuit. Find the Thévenin voltage and the Thévenin resistance. Form a diagram in the style of Figure 1a (copied below for reference).



- (b) You are given a voltage divider as shown below. Find R_1 and R_2 such that the Thévenin equivalent model is the same as that of (a). You are given that $V_S = 10V$.



3. NAND Circuit

Let us consider a NAND logic gate, as seen in Section 2. This circuit implements the boolean function $\overline{(A \cdot B)}$. The $\cdot$ stands for the AND operation, and the $\overline{}$ stands for NOT; combining them, we get NAND!

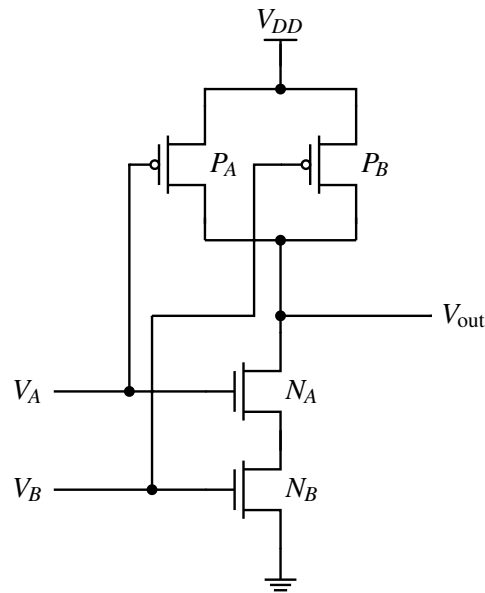


Figure 6: NAND gate transistor-level implementation.

V_{tn} and V_{tp} are the threshold voltages for the NMOS and PMOS transistors, respectively. Assume that $V_{DD} > V_{tn}$ and $|V_{tp}| > 0$.

- (a) Label the gate, source, and drain nodes for the NMOS and PMOS transistors above.

- (b) If $V_A = V_{DD}$ and $V_B = V_{DD}$, which transistors act like open switches? Which transistors act like closed switches? What is V_{out} ?

- (c) If $V_A = 0V$ and $V_B = V_{DD}$, what is V_{out} ?

- (d) If $V_A = V_{DD}$ and $V_B = 0V$, what is V_{out} ?

(e) If $V_A = 0V$ and $V_B = 0V$, what is V_{out} ?

(f) Write out the truth table for this circuit.

V_A	V_B	V_{out}
0	0	
0	V_{DD}	
V_{DD}	0	
V_{DD}	V_{DD}	

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Discussion Worksheet Discussion 2B

For this discussion, **Note 1** is helpful.

1. RC Circuits

In this problem, we will be using differential equations to find the voltage across a capacitor over time in an RC circuit. We set up our problem by first defining four functions over time: $I(t)$ is the current at time t , $V(t)$ is the voltage across the circuit at time t , $V_R(t)$ is the voltage across the resistor at time t , and $V_C(t)$ is the voltage across the capacitor at time t .

Recall from 16A that the voltage across a resistor is defined as $V_R = RI_R$ where I_R is the current across the resistor. Also, recall that the voltage across a capacitor is defined as $V_C = \frac{Q}{C}$ where Q is the charge across the capacitor.

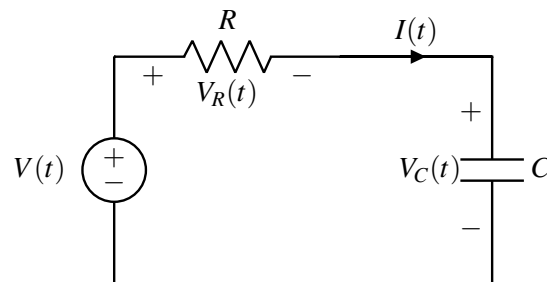


Figure 1: Example Circuit

- (a) First, find an equation that relates the current across the capacitor $I(t)$ with the voltage across the capacitor $V_C(t)$.

(b) Write a system of equations that relates the functions $I(t)$, $V_C(t)$, and $V(t)$.

(c) So far, we have three unknown functions and only one equation, but we can remove $I(t)$ from the equation using what we found in part (a). Rewrite the previous equation in part (b) in the form of a differential equation.

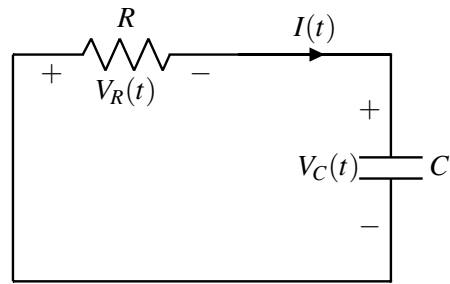


Figure 2: Circuit for part (d)

- (d) Let's suppose that at $t = 0$, the capacitor is charged to a voltage V_{DD} ($V_C(0) = V_{DD}$). Let's also assume that $V(t) = 0$ for all $t \geq 0$. Solve the differential equation for $V_C(t)$ for $t \geq 0$.

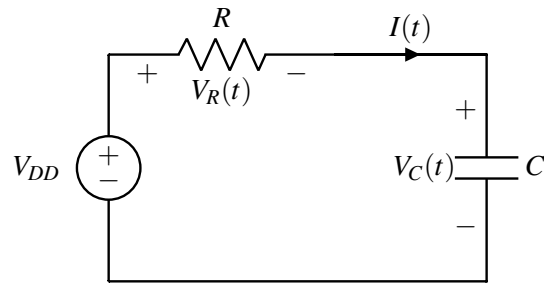


Figure 3: Circuit for part (e)

- (e) Now, let's suppose that we start with an uncharged capacitor $V_C(0) = 0$. We apply some constant voltage $V(t) = V_{DD}$ across the circuit. Solve the differential equation for $V_C(t)$ for $t \geq 0$.

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Discussion Worksheet Discussion 3A

For this discussion, **Note 2** is helpful.

1. Differential equations with piecewise constant inputs

Working through this question will help you understand better differential equations with inputs. Along the way, we will also touch a bit on going from continuous-time into a discrete-time view. This problem also provides a vehicle to review relevant concepts from calculus.

(a) Consider the scalar system

$$\frac{d}{dt}x(t) = \lambda x(t) + u(t). \quad (1)$$

Our goal is to solve this system (find an appropriate function $x(t)$) for general inputs $u(t)$. To do this, we will start with a piecewise constant $u(t)$; we already have the tools to solve this system, which we will do in the first few parts. Later in the worksheet, we will extend this to general $u(t)$.

Suppose that $x(t)$ is continuous (in real systems, this is almost always true). Further suppose that our input $u(t)$ of interest is piecewise constant over durations of width Δ . In other words:

$$u(t) = u(i\Delta) = u[i] \text{ if } t \in [i\Delta, (i+1)\Delta). \quad (2)$$

In keeping with this notation, we will use the notation

$$x_d[i] = x(i\Delta).$$

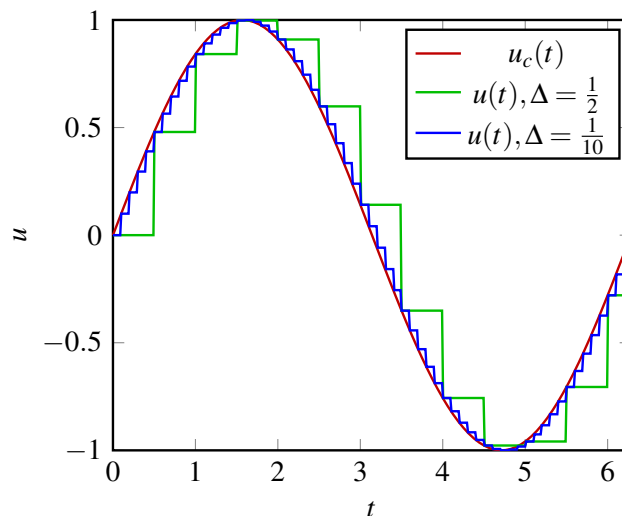


Figure 1: An example of a discrete input where the limit as the time-step Δ goes to 0 approaches a continuous function. The red line, the original signal $u_c(t) = \sin(t)$, is traced almost exactly by the blue line, which has a small time-step, and not nearly as well by the green line, which has a large time-step.

The first step to analyzing this system is to discover its behavior across a time-step with constant input, since we already know how to solve these kinds of systems.

Given that we know the value of $x(i\Delta) = x_d[i]$, compute $x_d[i+1] = x((i+1)\Delta)$.

Hint: For $t \in [i\Delta, (i+1)\Delta)$, the system is

$$\frac{d}{dt}x(t) = \lambda x(t) + u[i].$$

Also see [Note 2](#).

Here is a solution to this system, which may help with visual intuition:

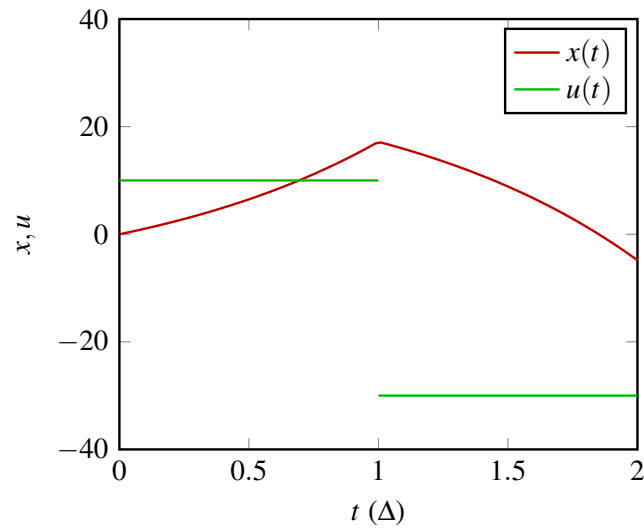


Figure 2: An example of a solution to this diff. eq. system. In this case $\lambda = 1, u[0] = 10, u[1] = -30$.

- (b) Now that we've found a one-step recurrence for $x_d[i+1]$ in terms of $x_d[i]$, we want to get an expression for $x_d[i]$ in terms of the original value $x(0) = x_d[0]$, and all the inputs u . This is so that we can eventually convert this function for $x_d[i]$ into a function for $x(t)$.

Unroll the implicit recursion you derived in the previous part to write $x_d[i+1]$ as a sum that involves $x_d[0]$ and the $u[j]$ for $j = 0, 1, \dots, i$.

For this part, feel free to just consider the discrete-time system in a simpler form

$$x_d[i+1] = ax_d[i] + bu[i] \quad (3)$$

and you don't need to worry about what a and b actually are in terms of λ and Δ .

(*Hint:* What is $x_d[1]$ in terms of $x_d[0]$? What is $x_d[2]$ in terms of (only) $x_d[0]$? What about $x_d[3]$? Can you find a pattern?)

- (c) For a given time t in continuous real time, what is the discrete i interval that corresponds to it?
(*Hint:* $\lfloor x \rfloor$ is the largest integer smaller than x .)

- (d) Here's the first payoff! Use the results of part (a) and (b) to give an approximate expression for $x(t)$ for any t , in terms of $x_d[0] = x(0)$ and the inputs $u[j]$. You can assume that Δ is small enough that $x(t)$ does not change too much (is approximately constant) over an interval of length Δ .

(Hint: The assumption we just made allows us to approximate $x(t) \approx x\left(\Delta \lfloor \frac{t}{\Delta} \rfloor\right) = x_d\left[\lfloor \frac{t}{\Delta} \rfloor\right]$.)

- (e) Now, we are going to turn this around. Suppose that the $u[i]$ is actually a sample of a desired input $u_c(t)$ in continuous time. Namely, suppose that $u[i] = u_c(i\Delta)$.

To clarify, $u(t)$ is a piecewise constant function; $u[i]$ is the discrete input that constructs $u(t)$; and $u_c(t)$ is the underlying input $u[i]$ is sampled from.

The underlying goal is to find an expression for $x(t)$ in the limit $\Delta \rightarrow 0$, in terms of $u_c(t)$ and the initial condition $x(0)$. To this end, start by substituting an appropriate value of u_c for u in the result from part (d). (Note: don't take any limits in this problem; just do the substitution.)

- (f) We want to take the limit $\Delta \rightarrow 0$ of our (discrete-time) expression and thus get a continuous-time function, but right now our discrete-time expression itself is pretty complicated. Let's simplify it by making some approximations which become exact in the limit.

Further approximate the previous expression by considering the following two estimates:

- i. Let $n = \lfloor \frac{t}{\Delta} \rfloor \approx \frac{t}{\Delta}$ where needed and treat $\Delta \approx \frac{t}{n}$. This is a meaningful approximation when we think about n large enough.
 - ii. Treat $\frac{1-e^{-\lambda\Delta}}{\lambda} \approx \Delta$. This is a meaningful approximation when we think about Δ small enough. One can derive this estimate by using Taylor's theorem from calculus, but it's not required here.
- (*Hint*: Use the first estimate to get rid of “floor” terms, then use both estimates to simplify further.)

- (g) Here's our second payoff! We now obtain a continuous-time expression for $x(t)$, completing the transition into continuous-time. Take the limit of $x(t)$ as $\Delta \rightarrow 0$ or equivalently as $n \rightarrow \infty$. What is the expression you get for $x(t)$?

(*Hint*: Remember your definition of definite integrals as limits of Riemann sums in calculus.)

(h) Verify the analytic solution for $x(t)$ found in (g) for $u(t) = 0$ and for $u(t) = u_0$.

(i) Verify the analytic solution found in (g) by plugging it back into the differential equation.

- (j) If input $u(t)$ is the linear sum of two other inputs $u(t) = c_1 u_1(t) + c_2 u_2(t)$, what does the solution look like? What does this mean?

This problem brings together many different concepts and uses a lot of notation. As such, it may be difficult to fully comprehend everything the first time. *It is perfectly fine* to go back and spend more time on the problem until you completely understand it. Being able to quickly analyze complex mathematical problems like this is part of the vaunted “mathematical maturity” that this class helps you foster. As the semester continues, you will find that these kinds of problems will seem progressively easier, both to understand quickly and to solve. But it won’t happen without practice.

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Discussion Worksheet Discussion 3B

The relevant notes for this discussion are the sections 1 and 2 of **Note 3**.

1. Changing Coordinates and Systems of Differential Equations

Suppose we have the pair of differential equations (valid for $t \geq 0$)

$$\frac{d}{dt}x_1(t) = -9x_1(t)$$

$$\frac{d}{dt}x_2(t) = -2x_2(t)$$

with initial conditions $x_1(0) = -1$ and $x_2(0) = 3$.

- (a) Solve for $x_1(t)$ and $x_2(t)$ for $t \geq 0$.

Suppose we are actually interested in a different set of variables with the following differential equations:

$$\begin{aligned}\frac{d}{dt}z_1(t) &= -5z_1(t) + 2z_2(t) \\ \frac{d}{dt}z_2(t) &= 6z_1(t) - 6z_2(t).\end{aligned}$$

- (b) Write out the above system of differential equations in matrix form. Assuming that the initial state $\vec{z}(0) = \begin{bmatrix} 7 & 7 \end{bmatrix}^T$, can we solve this system directly?

- (c) Consider that in our frustration with the previous system of differential equations, we start hearing voices. These voices whisper to us that that we should try the following change of variables:

$$\begin{aligned}z_1(t) &= -y_1(t) + 2y_2(t) \\ z_2(t) &= 2y_1(t) + 3y_2(t).\end{aligned}$$

Write out this transformation in matrix form ($\vec{z} = V\vec{y}$).

For each of the parts (d) - (f), solve the questions two ways: 1. using direct substitution, and 2. using matrices and vectors .

- (d) How do the initial conditions for $z_i(t)$ translate into the initial conditions for $y_i(t)$?
1. By direct substitution:

2. Using matrices and vectors:

(e) Rewrite the differential equations in terms of $y_i(t)$. Can we solve this system of differential equations?

1. By direct substitution:

2. Using matrices and vectors:

(f) What are the solutions for $z_i(t)$?

1. Solve this with direct substitution:

2. Solve this with matrices and vectors:

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 4A

The relevant notes for this discussion is **Note 3A**.

1. Changing Coordinates and Systems of Differential Equations, II

In the previous discussion we analyzed and solved a pair of differential equations where the variables of interest were coupled.

$$\begin{aligned}\frac{d}{dt}z_1(t) &= -5z_1(t) + 2z_2(t) \\ \frac{d}{dt}z_2(t) &= 6z_1(t) - 6z_2(t).\end{aligned}$$

We solved this system by using a coordinate transformation that gave us a decoupled system of equations. In the last discussion we were simply handed this transformation, but in this discussion we will construct the transformation for ourselves.

We will focus our explorations on the voltages across the capacitors in the following circuit.

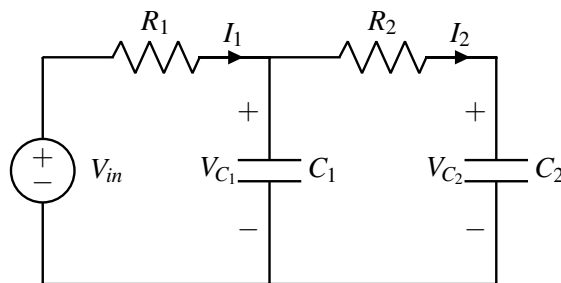


Figure 1: Two dimensional system: a circuit with two capacitors, like the one in lecture.

- (a) Write the system of differential equations governing the voltages across the capacitors V_{C_1}, V_{C_2} . Use the following values: $C_1 = 1\mu\text{F}, C_2 = \frac{1}{3}\mu\text{F}, R_1 = \frac{1}{3}\text{M}\Omega, R_2 = \frac{1}{2}\text{M}\Omega$.

- (b) Suppose also that V_{in} was at 7 Volts for a long time, and then transitioned to be 0 Volts at time $t = 0$. This "new" system of differential equations (valid for $t \geq 0$)

$$\frac{d}{dt}y_1(t) = -5y_1(t) + 2y_2(t) \quad (1)$$

$$\frac{d}{dt}y_2(t) = 6y_1(t) - 6y_2(t) \quad (2)$$

with initial conditions $y_1(0) = 7$ and $y_2(0) = 7$.

Write out the differential equations and initial conditions in matrix/vector form.

- (c) Find the eigenvalues λ_1 , λ_2 and eigenspaces for the matrix corresponding to the differential equation matrix above.

(d) Change coordinates into the eigenbasis to re-express the differential equations in terms of new variables $z_{\lambda_1}(t)$, $z_{\lambda_2}(t)$. (These variables represent eigenbasis-aligned coordinates.)

(e) Solve the differential equation for $z_{\lambda_i}(t)$ in the eigenbasis.

(f) Convert your solution back into the original coordinates to find $y_i(t)$.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 4B

In this discussion, we're going to cover linear algebra concepts (Change of Basis, Diagonalization) that unlock powerful circuit analysis techniques going forward. We also introduce a new kind of circuit element called the "inductor"; **Note 3B** will be useful.

1. Coordinate Change of Basis: Examples

Consider a vector in the standard basis,

$$\vec{x} = a\vec{e}_1 + b\vec{e}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \mathbf{I}\vec{x} \quad (1)$$

where, a, b are $\vec{x}$'s coordinates in the standard basis and $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are the elementary standard basis vectors.

Given a new set of basis vectors, $\mathcal{V} = \{\vec{v}_1, \vec{v}_2\}$, if $\vec{x} \in \text{span}\{\mathcal{V}\}$, then we can find new coordinates in terms of this new basis. The new coordinates are called a_v, b_v and are described,

$$\vec{x} = a_v\vec{v}_1 + b_v\vec{v}_2 = \begin{bmatrix} | & | \\ \vec{v}_1 & \vec{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} a_v \\ b_v \end{bmatrix} = \mathbf{V}\vec{x}_v \quad (2)$$

Now consider another set of basis vectors, $\mathcal{U} = \{\vec{u}_1, \vec{u}_2\}$. If $\vec{x} \in \text{span}\{\mathcal{U}\}$, then we can find the coordinates of $\vec{x}$ in terms of this basis. These coordinates are called a_u, b_u and are described,

$$\vec{x} = a_u\vec{u}_1 + b_u\vec{u}_2 = \begin{bmatrix} | & | \\ \vec{u}_1 & \vec{u}_2 \\ | & | \end{bmatrix} \begin{bmatrix} a_u \\ b_u \end{bmatrix} = \mathbf{U}\vec{x}_u \quad (3)$$

All of these bases are equivalent representations of any vector $\vec{x} \in \mathbb{R}^2$; each with their own set of coordinates. The same logic can, of course, be extended to any number of dimensions.

$$\vec{x} = \begin{bmatrix} | & | \\ \vec{e}_1 & \vec{e}_2 \\ | & | \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} | & | \\ \vec{u}_1 & \vec{u}_2 \\ | & | \end{bmatrix} \begin{bmatrix} a_u \\ b_u \end{bmatrix} = \begin{bmatrix} | & | \\ \vec{v}_1 & \vec{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} a_v \\ b_v \end{bmatrix} \quad (4)$$

$$\vec{x} = \mathbf{I}\vec{x} = \mathbf{V}\vec{x}_v = \mathbf{U}\vec{x}_u \quad (5)$$

Now that we've seen a conceptual overview of the change-of-basis, we can proceed with the worksheet problems.

(a) *Transformation From Standard Basis To Another Basis in $\mathbb{R}^3$*

Calculate the coordinate transformation between the following bases:

$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix},$$

i.e. find a matrix $\mathbf{T}$, such that $\vec{x}_v = \mathbf{T}\vec{x}_u$ where $\vec{x}_u$ contains the coordinates of a vector in a basis of the columns of $\mathbf{U}$ and $\vec{x}_v$ is the coordinates of the same vector in the basis of the columns of $\mathbf{V}$.

Let $\vec{x}_u = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and compute $\vec{x}_v$. Repeat this for $\vec{x}_u = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Now let $\vec{x}_u = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. What is $\vec{x}_v$?

(b) *Transformation Between Two Bases in $\mathbb{R}^3$*

Calculate the coordinate transformation between the following bases:

$$\mathbf{V} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix},$$

i.e. find a matrix $\mathbf{T}$, such that $\vec{x}_w = \mathbf{T}\vec{x}_v$. Let $\vec{x}_v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and compute $\vec{x}_w$. Repeat this for $\vec{x}_v = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

Now let $\vec{x}_v = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$. What is $\vec{x}_w$?

2. Diagonalization

- (a) Consider a matrix $\mathbf{A}$, a matrix $\mathbf{V}$ whose columns are the eigenvectors of $\mathbf{A}$, and a diagonal matrix $\mathbf{\Lambda}$ with the eigenvalues of $\mathbf{A}$ on the diagonal (in the same order as the eigenvectors (or columns) of $\mathbf{V}$). From these definitions, show that

$$\mathbf{AV} = \mathbf{V\Lambda}$$

3. Introduction to Inductors

An inductor is a circuit element analogous to a capacitor; its voltage changes as a function of the derivative of the current across it. That is:

$$V_L(t) = L \frac{dI_L(t)}{dt}$$

When first studying capacitors, we analyzed a circuit where a current source was directly attached to a capacitor. In Figure 1, we form the equivalent "fundamental" circuit for an inductor:

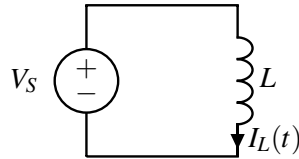


Figure 1: Inductor in series with a voltage source.

- (a) What is the current through an inductor as a function of time? If the inductance is $L = 3\text{H}$, what is the current at $t = 6\text{s}$? Assume that the voltage source turns from 0V to 5V at time $t = 0\text{s}$, and there's no current flowing in the circuit before the voltage source turns on.

- (b) Now, we add some resistance in series with the inductor, as in Figure 2.

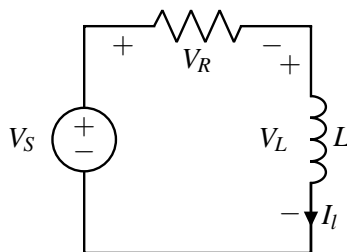


Figure 2: Inductor in series with a voltage source.

Solve for the current $I_L(t)$ in the circuit over time, in terms of R, L, V_S, t .

- (c) **(Practice)** Suppose $R = 500\Omega$, $L = 1\text{mH}$, $V_S = 5\text{V}$. Plot the current through and voltage across the inductor ($I_L(t)$, $V_L(t)$), as these quantities evolve over time.

4. Fibonacci Sequence

- (a) The Fibonacci sequence is built as follows: the n -th number (F_n) is sum of the previous two numbers in the sequence. That is:

$$F_n = F_{n-1} + F_{n-2}$$

If the sequence is initialized with $F_1 = 0$ and $F_2 = 1$, then the first 11 numbers in the Fibonacci sequence are:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

We can express this computation as a matrix multiplication:

$$\begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} = \mathbf{A} \begin{bmatrix} F_{n-1} \\ F_{n-2} \end{bmatrix}$$

What is $\mathbf{A}$?

- (b) Find the eigenvalues and corresponding eigenvectors of $\mathbf{A}$.

(c) Diagonalize $\mathbf{A}$ (that is, in the expression $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, solve for each component matrix.)

(d) Use the diagonalized result to show that we can arrive at an analytical result for any F_n :

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n-1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n-1}$$

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 5B

For this discussion, **Note 4** is helpful.

Some equations which may be useful are:

- $\sin(x) = \cos\left(x - \frac{\pi}{2}\right)$ for all real numbers x .
- $\cos(x) = \frac{1}{2}(e^{jx} + e^{-jx})$ for all real numbers x .
- The impedance of a resistor with resistance R is $Z_R = R$.
- The impedance of a capacitor with capacitance C is $Z_C = \frac{1}{j\omega C}$.
- The impedance of an inductor with inductance L is $Z_L = j\omega L$.

1. Phasor Analysis

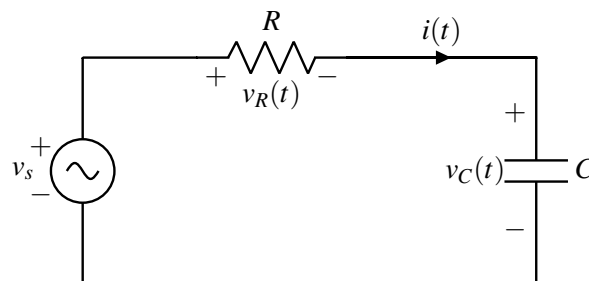
Any sinusoidal time-varying function $x(t)$, representing a voltage or a current, can be expressed in the form

$$x(t) = \tilde{X}e^{j\omega t} + \tilde{\tilde{X}}e^{-j\omega t}, \quad (1)$$

where $\tilde{X}$ is a time-independent function called the **phasor** representation of $x(t)$ (recall that $\bar{a}$ denotes the complex conjugate of a). Note that 1) $\tilde{X}$ and $\tilde{\tilde{X}}$ are complex conjugates of each other, 2) $e^{j\omega t}$ and $e^{-j\omega t}$ are complex conjugates of each other, and 3) that $\tilde{X}e^{j\omega t}$ and $\tilde{\tilde{X}}e^{-j\omega t}$ are also complex conjugates of each other.

Note: We define the phasor as the coefficient of $e^{j\omega t}$ in eq. (1). Other resources (such as some past iterations of this class) define it slightly differently; the definitions differ by a constant multiple. Some reasons for competing definitions are discussed in **Note 4**. Although the definitions in general lead to the same answers, be careful to use our class' definition and not get tripped up. For example, if we ask about the magnitude of the phasor, you wouldn't want to be off by a constant!

The phasor analysis method consists of five steps. Consider the RC circuit below.



The voltage source is given by

$$v_s(t) = 12 \sin\left(\omega t - \frac{\pi}{4}\right), \quad (2)$$

with $\omega = 1 \times 10^3 \frac{\text{rad}}{\text{s}}$, $R = \sqrt{3}\text{k}\Omega$, and $C = 1\text{ }\mu\text{F}$.

Our goal is to obtain a solution for $i(t)$ with the sinusoidal voltage source $v_s(t)$.

(a) **Step 1: Write sources as exponentials:** $\tilde{X}e^{j\omega t} + \tilde{X}e^{-j\omega t}$

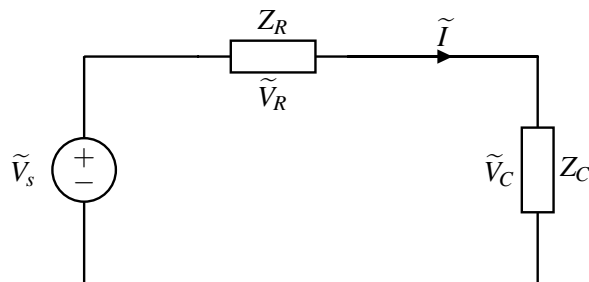
All voltages and currents with known sinusoidal functions should be expressed in the standard exponential format. Convert $v_s(t)$ into an exponential and write down its phasor representation $\tilde{V}_s$.

(b) **Step 2: Transform circuits to phasor domain**

The voltage source is represented by its phasor $\tilde{V}_s$. The current $i(t)$ is related to its phasor counterpart $\tilde{I}$ by

$$i(t) = \tilde{I}e^{j\omega t} + \tilde{I}e^{-j\omega t}.$$

We redraw the circuit in phasor domain:



What are the impedances of the resistor, Z_R , and capacitor, Z_C ? We sometimes also refer to this as the "phasor representation" of R and C .

(c) **Step 3: Cast the branch and element equations in phasor domain**

Use Kirchhoff's laws to write down a loop equation that relates all phasors in Step 2.

(d) **Step 4: Solve for unknown variables**

Solve the equation you derived in Step 3 for $\tilde{I}$ and $\tilde{V}_C$. What is the polar form of $\tilde{I}$ and $\tilde{V}_C$? Polar form is given by $Ae^{j\theta}$, where A is a positive real number.

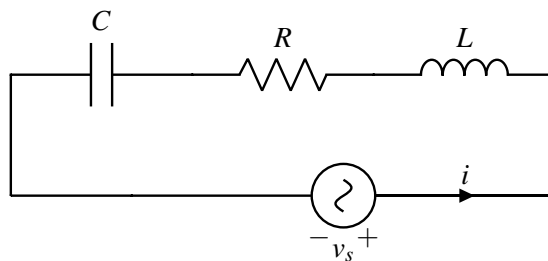
(e) Step 5: Transform solutions back to time domain

To return to time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart. What is $i(t)$ and $v_C(t)$? What is the phase difference between $i(t)$ and $v_C(t)$?

2. RLC Circuit Phasor Analysis

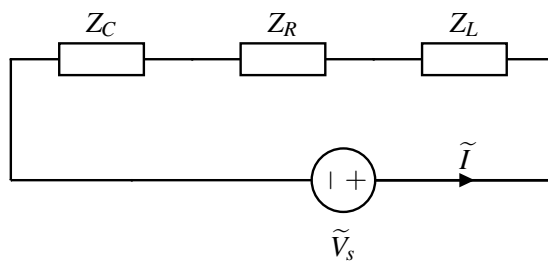
We study a simple RLC circuit with an AC voltage source given by

$$v_s(t) = B \cos(\omega t - \phi)$$



(a) Write out the phasor representation of $v_s(t)$, and the impedances of R , C , and L .

(b) Now, we're going to redraw the circuit in the phasor domain.



Use Kirchhoff's laws to write down a loop equation relating the phasors.

(c) Solve the equation in the previous step for the current $\tilde{I}$. What is the magnitude and phase of the polar form of $\tilde{I}$?

Hint: You'll need the following identities, which you can find in [Note j](#):

- $|z_1/z_2| = |z_1|/|z_2|$
- $\angle\left(\frac{z_1}{z_2}\right) = \angle z_1 - \angle z_2$
- $\angle(a + jb) = \text{atan2}(b, a)$ ¹

¹We use $\text{atan2}(b, a)$ because passing in $\frac{b}{a}$ to the $\tan^{-1}$ or arctangent function loses information about the signs of b and a , and thus the location of $a + jb$. This is the reason $\tan^{-1}$ outputs numbers only in the range $[-\frac{\pi}{2}, \frac{\pi}{2}]$; on the other hand, atan2 outputs numbers in the range $[0, 2\pi]$, which is what we want. More information about atan2 is found in [Note j](#).

3. (Practice) Inductor Impedance

Given the voltage-current relationship of an inductor $v(t) = L \frac{di(t)}{dt}$, show that its complex impedance is $Z_L = j\omega L$.

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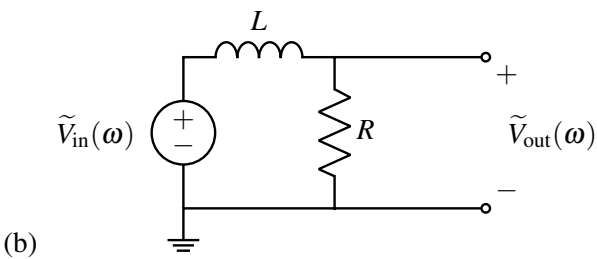
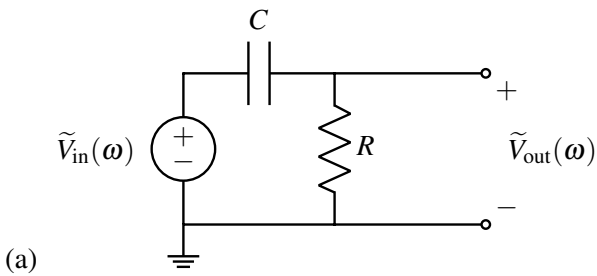
EECS 16B Designing Information Devices and Systems II

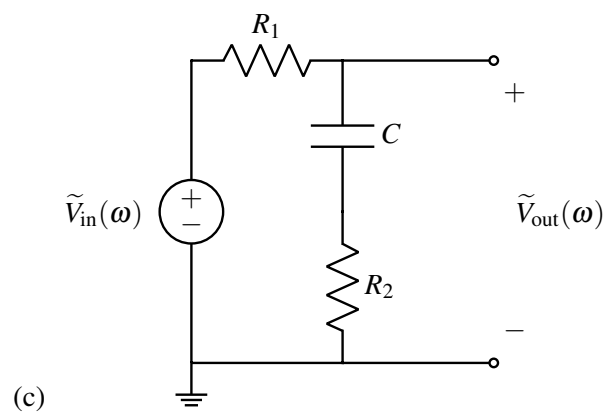
Discussion Worksheet Discussion 6A

The relevant notes for this discussion are **Note 5** and **Note 6**.

1. Transfer function practice

In this problem, you'll be deriving some transfer functions on your own. For each circuit, determine $H(\omega) = \frac{\tilde{V}_{\text{out}}(\omega)}{\tilde{V}_{\text{in}}(\omega)}$. How does each circuit respond as $\omega \rightarrow 0$ (low frequencies)? as $\omega \rightarrow \infty$ (high frequencies)?





2. Plotting and combining transfer functions

Recall that any transfer function can be written in polar form as

$$H(\omega) = A(\omega)e^{j\alpha(\omega)}$$

where $A(\omega)$ and $\alpha(\omega)$ are real functions of omega giving the magnitude and phase of the transfer function, respectively. To see how transfer functions combine, consider two $H_1(\omega)$ and $H_2(\omega)$.

$$H_1(\omega) = Ae^{j\alpha}$$

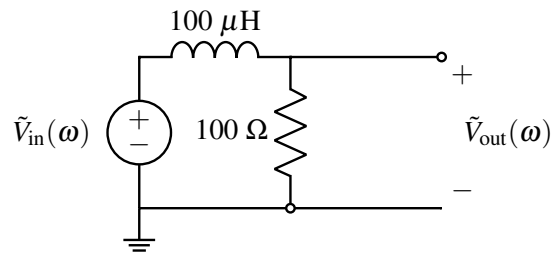
$$H_2(\omega) = Be^{j\beta}$$

$$H_1(\omega) \cdot H_2(\omega) = Ae^{j\alpha}Be^{j\beta} = AB e^{j(\alpha+\beta)}$$

$$\frac{H_1(\omega)}{H_2(\omega)} = \frac{Ae^{j\alpha}}{Be^{j\beta}} = \frac{A}{B} e^{j(\alpha-\beta)}$$

As you can see, magnitudes of transfer functions multiply and divide while the phases add and subtract.

In this problem we will try to plot the transfer function of the following circuit:



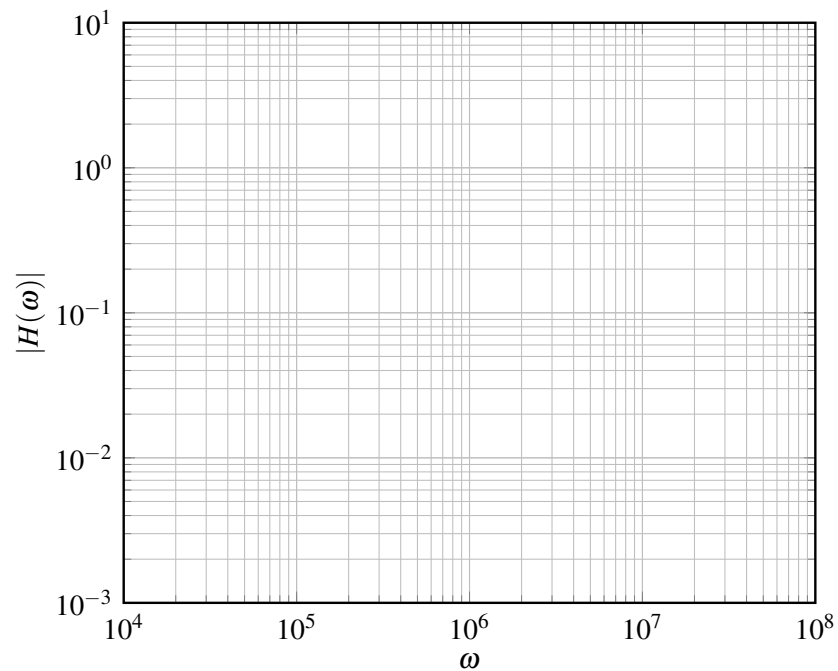
(a) Write expressions for $|H(\omega)|$ and $\angle H(\omega)$. For now, you can keep it in terms of R and L .

(b) What is the cutoff frequency for this circuit? Mark it on the log-log plot with a vertical line. Recall that a transfer function of the form $H(\omega) = \frac{k}{1+j\omega/\omega_c}$ has a cutoff frequency of ω_c .

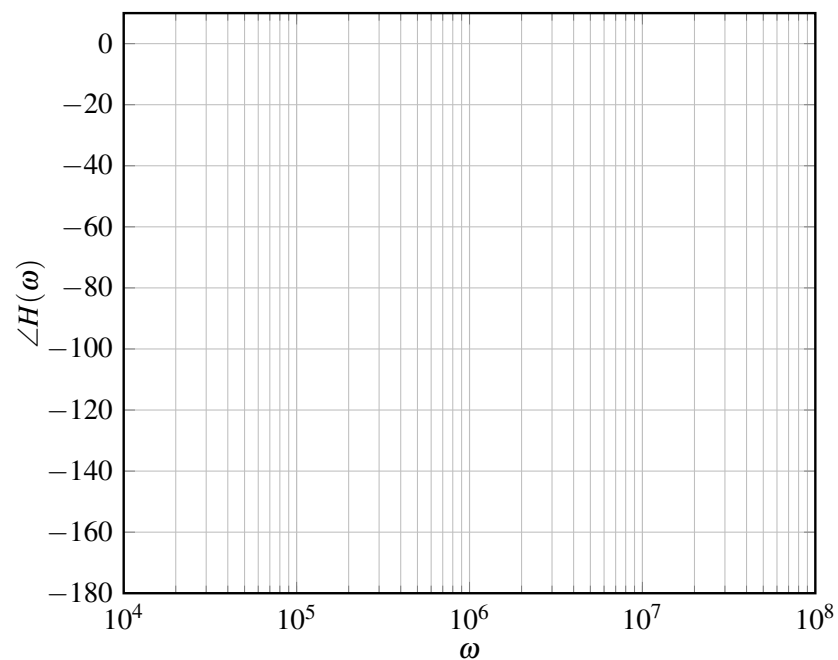
- (c) Sketch plots of the magnitude and phase of this transfer function. We have provided a table with the transfer function evaluated at a few representative points around the cutoff frequency.

ω	10^4	10^5	10^6	10^7	10^8
$ H(\omega) $	1.00	0.995	0.707	0.100	0.01
$\angle H(\omega)$	-0.6	-6	-45	-84	-89

Plot of $|H(\omega)|$ (for **you** to draw).



Plot of $\angle H(\omega)$ (for **you** to draw).

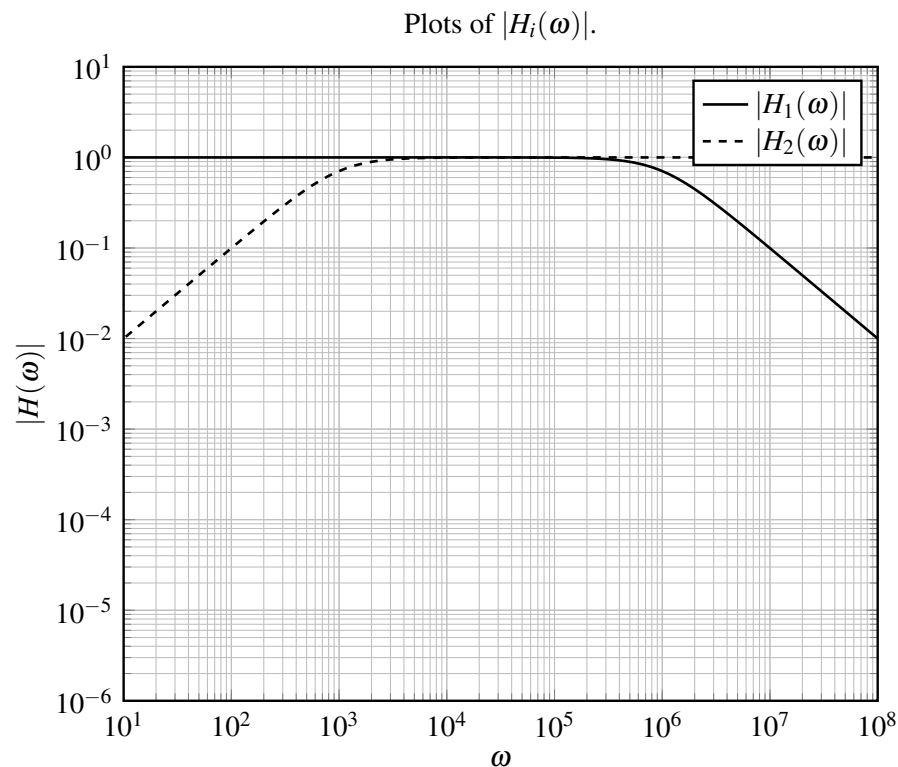


- (d) Now suppose we want to compose the filter from part (1.a) with this new circuit. Use $R = 1\text{k}\Omega$ and $C = 1\mu\text{F}$ for the filter from part (1.a). We can compose two circuits by connecting the output of the first circuit into the second circuit, through a unity gain buffer. For this problem, call the circuit from this system with an inductor H_1 , and the filter from (1.a) H_2 . The transfer function of the composed circuit is:

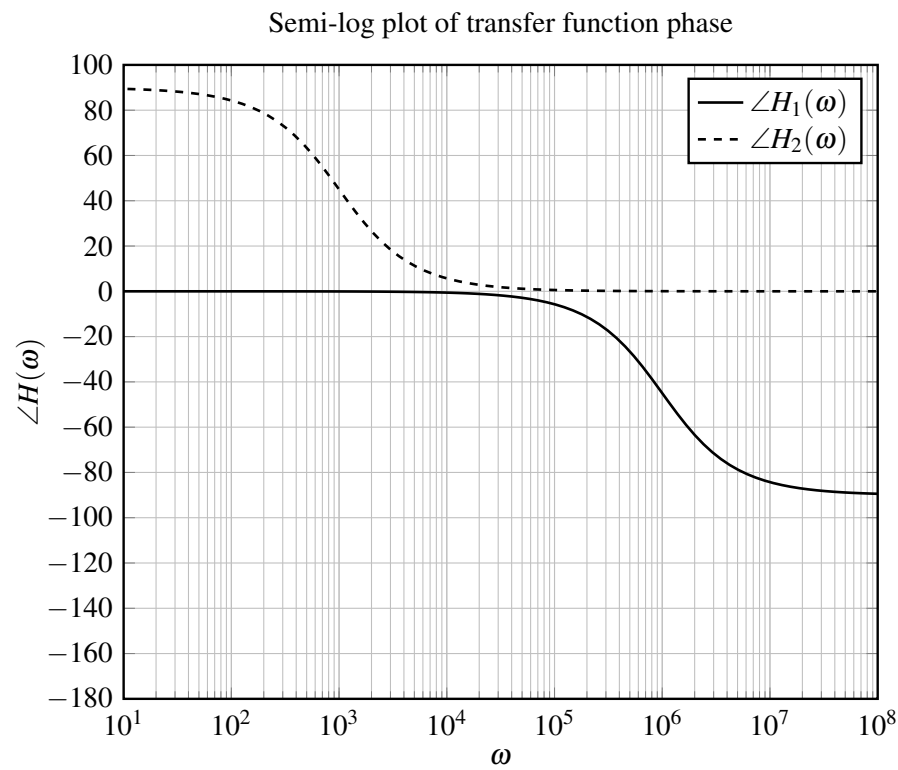
$$H(\omega) = H_1(\omega) \cdot H_2(\omega)$$

- i. Draw this circuit.

- ii. Plot the magnitude of the composed circuit. Below is a log-log plot with the magnitudes of $|H_1(\omega)|$ and $|H_2(\omega)|$ drawn to assist you. Recall that $|AB| = |A| \cdot |B|$ for complex numbers A and B .



- ii. Plot the phase of the composed circuit. Below is a semi-log plot with the phases $\angle H_1(\omega)$ and $\angle H_2(\omega)$ drawn to assist you.



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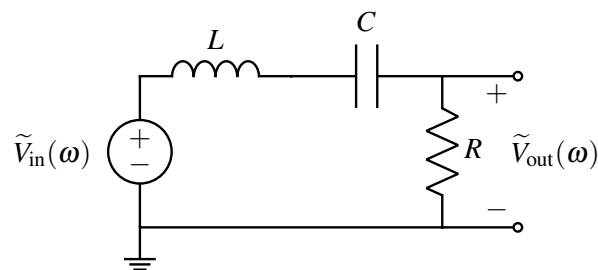
EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 6B

For this discussion, **Note 5** and **Note 6** are helpful.

1. Band-pass filter

It is quite common to need to design a filter which selects only a narrow range of frequencies. One example is in WiFi radios, it is desirable to select only the 2.4GHz frequency containing your data, and reject information from other nearby cellular or bluetooth frequencies. This type of filter is called a band-pass filter; we will explore the design of this type of filter in this problem.



- (a) **Write down the impedance of the series RLC combination in the form $Z_{RLC}(\omega) = A(\omega) + jX(\omega)$, where $A(\omega)$ and $X(\omega)$ are real valued functions of ω .**

- (b) **Write down the transfer function $H(\omega) = \frac{\tilde{V}_{out}(\omega)}{\tilde{V}_{in}(\omega)}$ for this circuit.**

- (c) **At what frequency ω_n does $X(\omega_n) = 0$?** (i.e. at what frequency is the impedance of the series combination of RLC purely real — meaning that the imaginary terms coming from the capacitor and inductor completely cancel each other. This is called the *natural frequency*.)

What happens to the relative magnitude of the impedances of the capacitor and inductor as ω moves above and below ω_n ? What is the value of the transfer function at this frequency ω_n ?

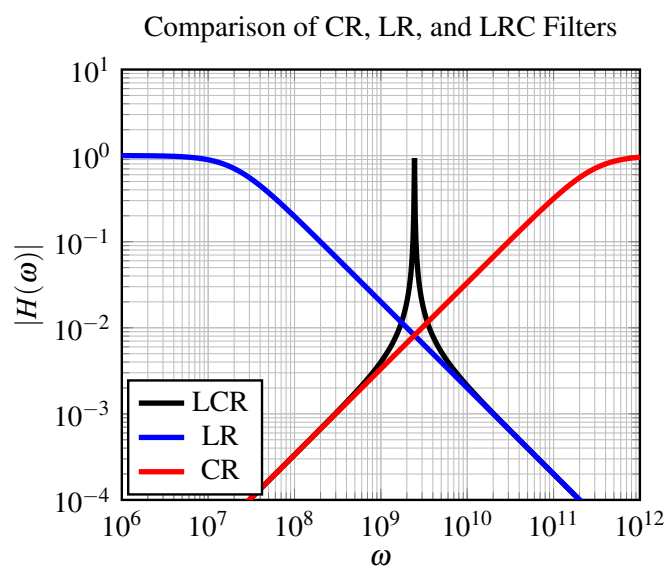
- (d) In most filters, we are interested in the cutoff frequency, since that helps define the frequency range over which the filter operates. Remember that this is the frequency at which the magnitude of the transfer function drops by a factor of $\sqrt{2}$ from its maximum value. Recalling that $H(\omega)$ can be written in the form $H(\omega) = \frac{R}{R + jX(\omega)}$, so that $|H(\omega)| = \frac{R}{\sqrt{R^2 + X(\omega)^2}}$, we see that $|H(\omega)| \leq 1$. Thus, the cutoff frequency is ω_c such that $|H(\omega_c)| = \frac{1}{\sqrt{2}}$, which is when $X(\omega_c) = \pm R$.

We want to find such ω_c . To do this, we want to see what happens in the neighborhood of ω_n and so express the combined effect of the capacitor and inductor in terms of $\omega = \omega_n + \Delta\omega$, where $\Delta\omega$ is (presumably) a relatively small number compared to ω_n .

Write an expression for $X(\omega_n + \Delta\omega)$, where $\Delta\omega$ is a variable shift from ω_n . Find the values of $\Delta\omega_1$ and $\Delta\omega_2$ which give $X(\omega_n + \Delta\omega_1) = -R$, and $X(\omega_n + \Delta\omega_2) = +R$. You may use the approximation that $\frac{1}{1+x} \approx 1 - x$ if $x \ll 1$.

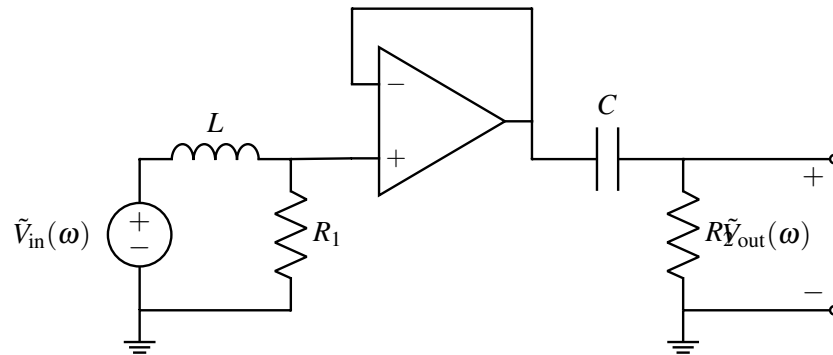
- (e) **Simplify $X(\omega)$ in two cases, when $\omega \rightarrow \infty$ and when $\omega \rightarrow 0$. Plug this simplified $X(\omega)$ into your previously solved expressions to find the transfer function at high and low frequencies.**

Our band-pass filter looks like an LR low-pass filter at high frequencies and a CR high-pass filter at low frequencies. Note that in this case, the cutoff frequencies for the LR and CR filters are not the same as LCR cutoff frequencies or the resonance frequency. In the vicinity of the resonance frequency, notice that the filter is much sharper than any first-order LR or RC filter.



2. Bode Plots

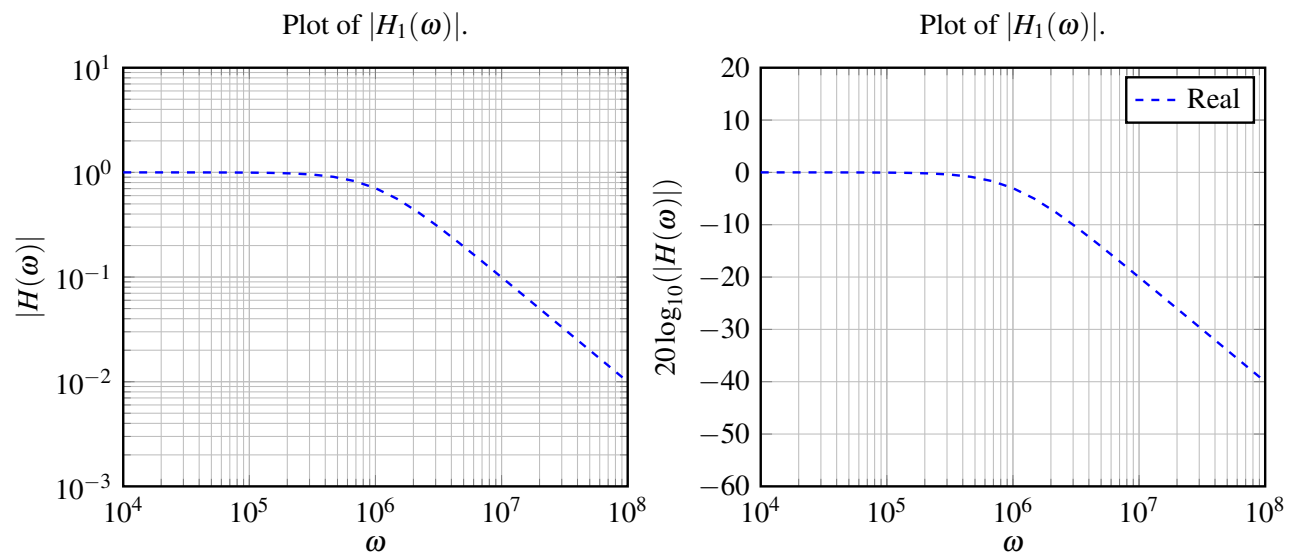
Our eventual goal is to construct Bode plots of the following circuit, with $L = 100\mu\text{H}$, $C = 1\mu\text{F}$, $R_1 = 100\Omega$, and $R_2 = 1\text{k}\Omega$:



To do this we will leverage the fact that Bode plots can be composed in simple ways.

Before we start diving into the problem, let's consider a modification of the *magnitude* plot that will help us work with multiple magnitude plots at once. Namely, instead of plotting $|H(\omega)|$ vs. ω where the y-axis is on a *logarithmic* scale, we plot $20\log_{10}(|H(\omega)|)$ vs. ω instead, and now the y-axis is on a *linear* scale.

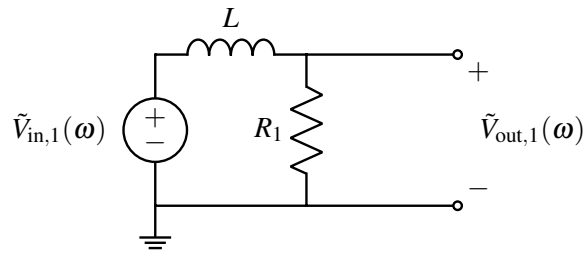
The reason that we do this is that, when combining transfer functions, we end up multiplying them. But we really want to add two plots graphically, not multiply them, so we will just plot and add the logarithms. (The constant multiple 20 is there for convention reasons.) Here's what this looks like, with the old grid on the left, and the new grid on the right:



Notice that we do not need to do this for the *phase* plots, since their y axes are naturally in linear scale, and combining plots can already be done by addition.

Now we are ready to begin working on the problems.

- (a) Consider the first half of this circuit:



We learned in the previous discussion that the transfer function is given by

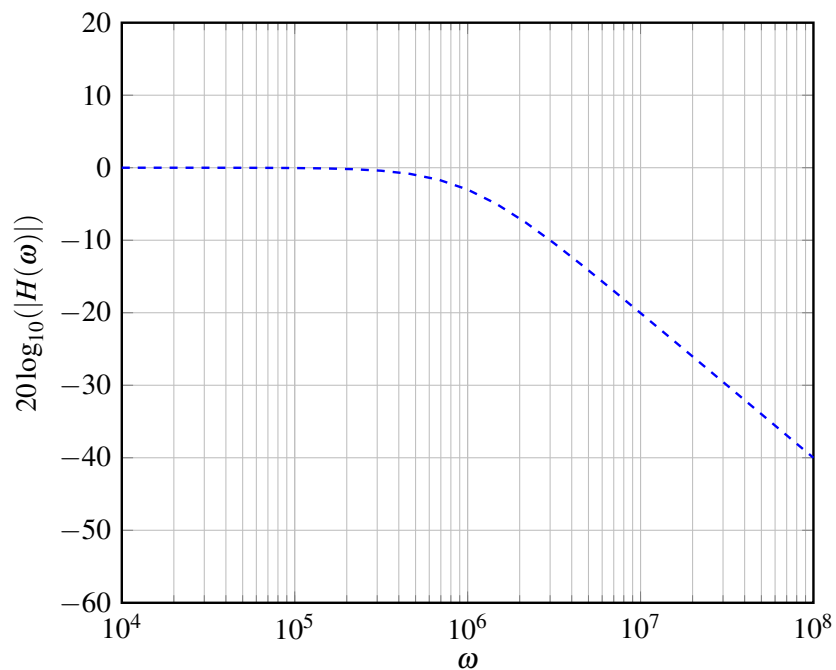
$$H_1(\omega) = \frac{\tilde{V}_{out,1}}{\tilde{V}_{in,1}} = \frac{1}{1 + j\omega \frac{L}{R_1}},$$

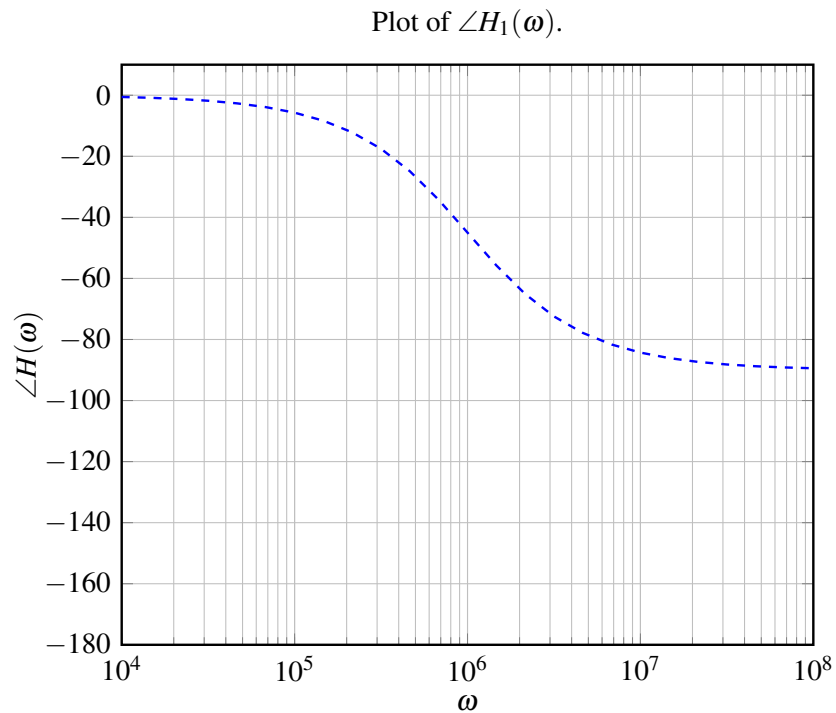
the cutoff frequency $\omega_{c,1}$ is given by

$$\omega_{c,1} = \frac{R_1}{L} = \frac{100\Omega}{100\mu\text{H}} = 1 \times 10^6 \frac{\text{rad}}{\text{s}},$$

and plots of the transfer function are given by

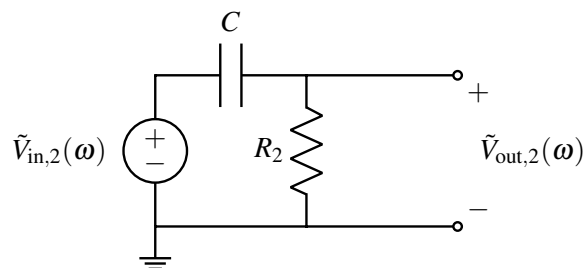
Plot of $|H_1(\omega)|$.





On these grids, **draw the Bode plots for magnitude and phase.**

(b) Consider the second half of the circuit:



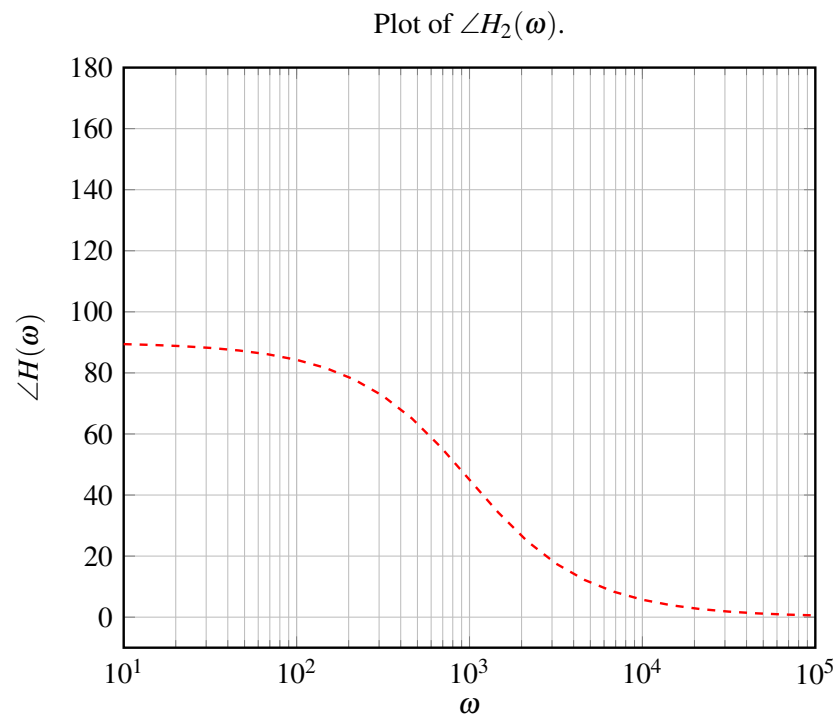
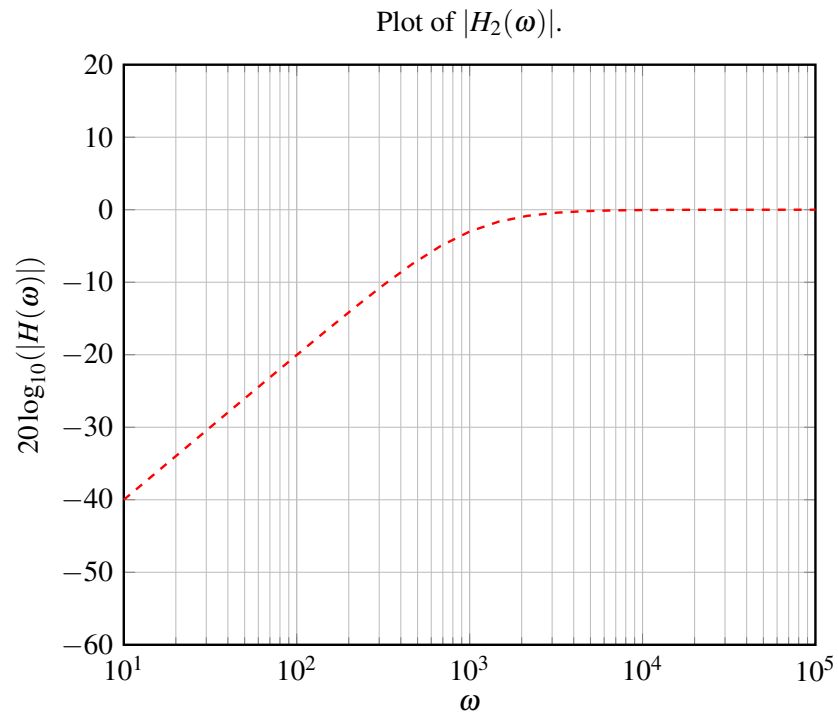
We learned in the previous discussion that the transfer function is given by

$$H_2(\omega) = \frac{\tilde{V}_{out,2}}{\tilde{V}_{in,2}} = \frac{j\omega R_2 C}{1 + j\omega R_2 C},$$

the cutoff frequency $\omega_{c,2}$ is given by

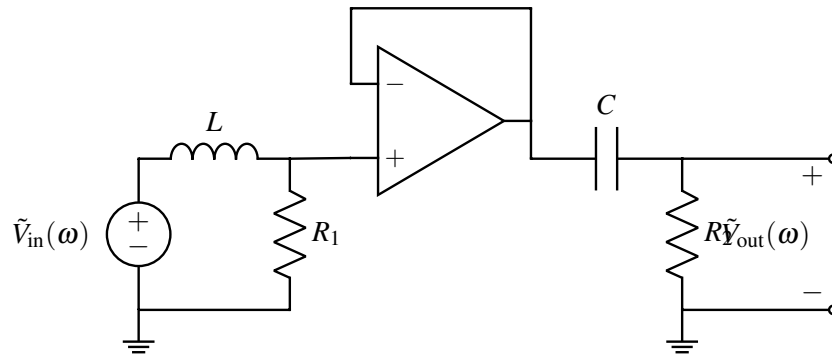
$$\omega_{c,2} = \frac{1}{R_2 C} = \frac{1}{(1\text{ k}\Omega) \cdot (1\text{ }\mu\text{F})} = 1 \times 10^3 \frac{\text{rad}}{\text{s}},$$

and plots of the transfer function are given by



On these grids, **draw the Bode plots for magnitude and phase.**

(c) Now, we will put this circuit together. Recall the original diagram:

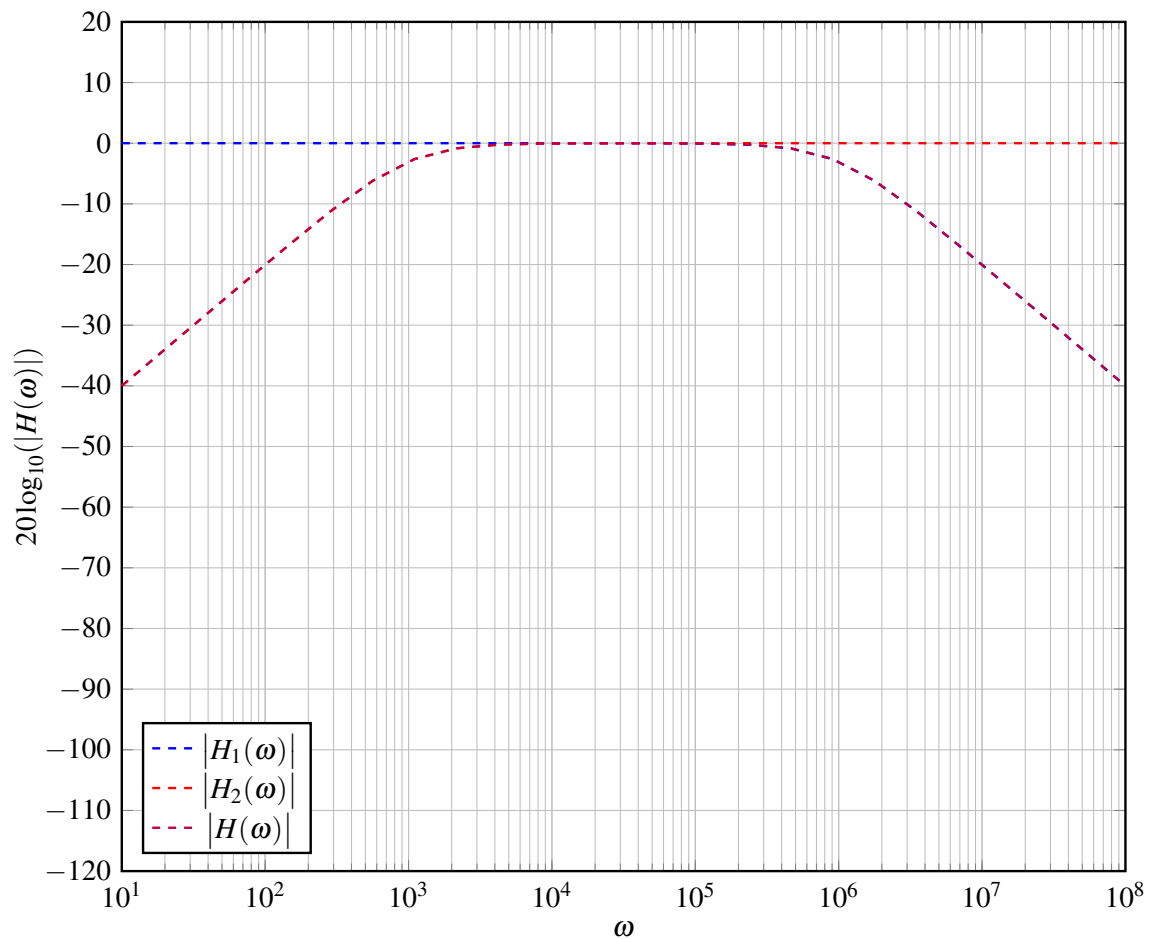


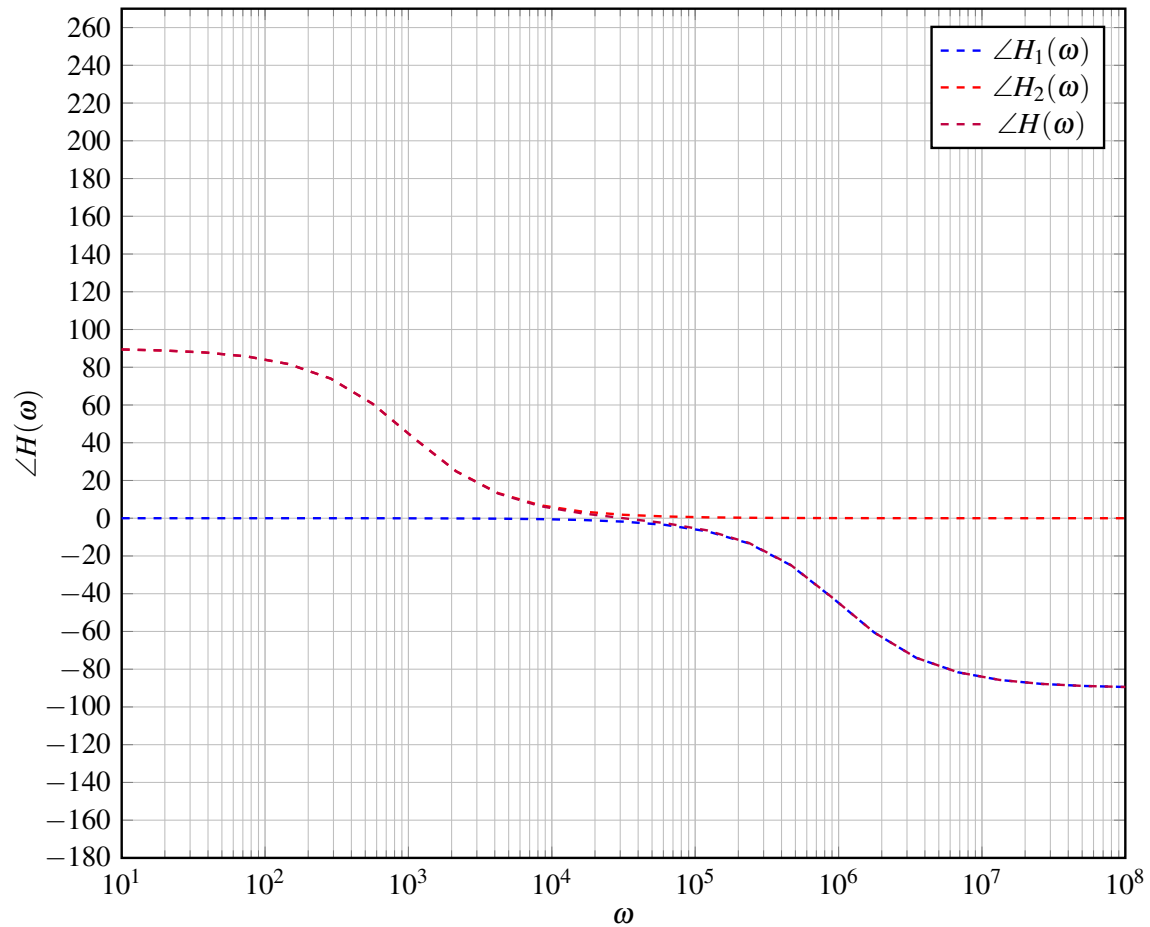
We learned in the previous discussion that the transfer function is

$$H(\omega) = \frac{\tilde{V}_{out}}{\tilde{V}_{in}} = H_1(\omega)H_2(\omega)$$

and the transfer function plots are given by

Plot of $|H(\omega)|$.



Plot of $\angle H(\omega)$.

On these grids, **draw the Bode plots for magnitude and phase.**

Hint: Recall that

$$\begin{aligned}
 20\log_{10}(|H(\omega)|) &= 20\log_{10}(|H_1(\omega)H_2(\omega)|) = 20\log_{10}(|H_1(\omega)|) + 20\log_{10}(|H_2(\omega)|) \\
 &= 20\log_{10}(|H_1(\omega)|) + 20\log_{10}(|H_2(\omega)|) \\
 \text{and } \angle H(\omega) &= \angle H_1(\omega) + \angle H_2(\omega).
 \end{aligned}$$

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 7A

The first question in this discussion clarifies some results cited in [Note 7A](#) and extends the results from Thursday's lecture on Discretization. The second question discusses system evolution over time, and sets the stage for discussing system stability.

1. A System Governed by Differential Equations: Piecewise Constant Inputs

Working through this question will help you better understand differential equations with inputs, and the sampling of a continuous-time system of differential equations into a discrete-time view. These concepts are important for control, since it is often easier to think about doing what we want in discrete-time. This question should initially feel similar to [dis03A](#), and in later subparts, we extend our analysis to the case of a vector differential equation.

(a) Consider the scalar system below:

$$\frac{d}{dt}x(t) = \lambda x(t) + u(t). \quad (1)$$

Further suppose that our input $u(t)$ of interest is piecewise constant over durations of width Δ . This is the same case we considered in [dis03A](#). In other words:

$$u(t) = u(i\Delta) = u_d[i] \text{ if } t \in [i\Delta, (i+1)\Delta). \quad (2)$$

Similarly, for $x(t)$.

$$x_d[i] = x(i\Delta).$$

Given that we start at $x(i\Delta)$, where do we end up at $x((i+1)\Delta)$?

- (b) Now that we've found a one-step recurrence for $x_d[i+1]$ in terms of $x_d[i]$, we want to get an expression for $x_d[i]$ in terms of the original value $x(0) = x_d[0]$, and all the inputs u . This is so that we can eventually convert this function for $x_d[i]$ into a function for $x(t)$.

Unroll the implicit recursion you derived in the previous part to write $x_d[i+1]$ as a sum that involves $x_d[0]$ and the $u_d[j]$ for $j = 0, 1, \dots, i$. The idea is to express the value of the discrete system at any arbitrary time solely in terms of where it started, and the accumulating set of inputs until that time.

For this part, to lighten the notation, feel free to just consider the discrete-time system in a simpler form

$$x_d[i+1] = ax_d[i] + bu_d[i] \quad (3)$$

and you don't need to worry about what a and b actually are in terms of λ and Δ .

- (c) Suppose we now have a continuous-time system of differential equations, that forms a vector differential equation. We express this with an input in vector form:

$$\frac{d}{dt}\vec{x}(t) = A\vec{x}(t) + \vec{b}u(t)$$

where $\vec{x}(t)$ is n -dimensional. We will assume that from here onwards, $\Delta = 1$, so that $i\Delta = i$.

Suppose further that the matrix A has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, with corresponding eigenvectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$. We collect the eigenvectors together and form the matrix $V = [\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n]$. (*Hint: What's the significance of this information?*)

If we apply a piecewise constant control input $u_d[i]$ as in (2), and sample the system $\vec{x}(t)$ at time intervals $t = i$, what are the corresponding A_d and $\vec{b}_d$ in:

$$\vec{x}((i+1)\Delta) \equiv \vec{x}(i+1) = A_d\vec{x}(i) + \vec{b}_d u_d[i] \quad (4)$$

- (d) We can write the relationship between the continuous and discrete systems as $\vec{x}(i) = \vec{x}_d[i]$, specifically since $\Delta = 1$. Suppose that $\vec{x}_d[0] = \vec{x}_0$. In the style of part b), unroll the implicit recursion you derived in the previous part to write $\vec{x}_d[i+1]$ as a sum that involves $\vec{x}_0$ and the $u_d[j]$ for $j = 0, 1, \dots, i$.

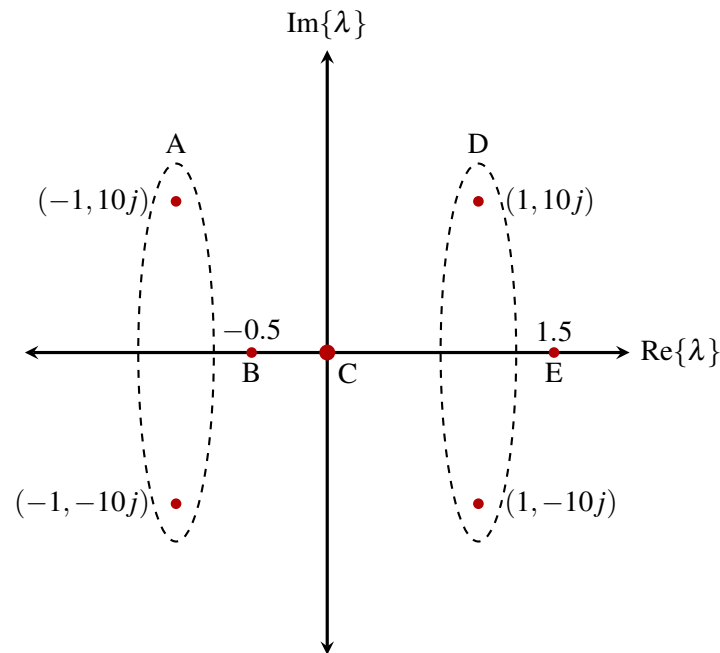
For this part, feel free to just consider the discrete-time system in the simpler form

$$\vec{x}_d[i+1] = A_d\vec{x}_d[i] + \vec{b}_d u_d[i] \quad (5)$$

and you don't need to worry about what A_d and $\vec{b}_d$ actually are in terms of the original parameters.

2. Continuous-time System Responses

We have a differential equation $\frac{d\vec{x}(t)}{dt} = A\vec{x}(t)$, where A is diagonal and has eigenvalues λ . For systems A, D, which have more than 1 eigenvalue, this equation is a vector differential equation; in the other cases (B, C, E), it is a scalar differential equation. For each set of λ values plotted on the real-imaginary axis, sketch $\vec{x}_1(t)$ with an initial condition of $\vec{x}_1(0) = 1$. In the scalar case, $\vec{x}_1(t) \equiv x_1(t) \equiv x(t)$.



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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 7B

The relevant notes for this discussion are [Note 7B](#) and [Note 8](#).

1. System identification by means of least squares

Working through this question will help you understand better how we can use experimental data taken from a (presumably) linear system to learn a discrete-time linear model for it using the least-squares techniques you learned in 16A. You will later do this in lab for your robot car.

As you were told in 16A, least-squares and its variants are not just the basic workhorses of machine learning in practice, they play a conceptually central place in our understanding of machine learning well beyond least-squares.

Throughout this question, you should consider measurements to have been taken from one long trace through time.

- (a) Consider the scalar discrete-time system

$$x[i+1] = ax[i] + bu[i] + w[i] \quad (1)$$

Where the scalar state at time i is $x[i]$, the input applied at time i is $u[i]$ and $w[i]$ represents some external disturbance that also participated at time i .

Assume that you have measurements for the states $x[i]$ from $i = 0$ to m and also measurements for the controls $u[i]$ from $i = 0$ to $m - 1$.

Set up a least-squares problem that you can solve to get an estimate of the unknown system parameters a and b .

- (b) What if there were now two distinct scalar inputs to a scalar system

$$x[i+1] = ax[i] + b_1u_1[i] + b_2u_2[i] + w[i] \quad (2)$$

and that we have measurements as before, but now also for both of the control inputs.

Set up a least-squares problem that you can solve to get an estimate of the unknown system parameters a, b_1, b_2 .

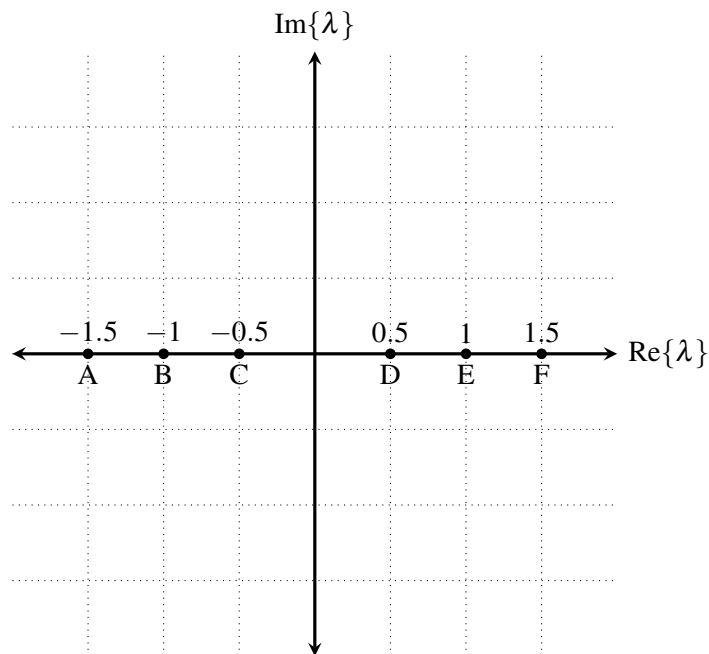
- (c) What could go wrong in the previous case? For what kind of inputs would make least-squares fail to give you the parameters you want?
- (d) Returning to the scalar case with only one input, what could go wrong? When would you be unable to use least-squares to get the parameters you want?
- (e) Now consider the two dimensional state case with a single input.

$$\vec{x}[i+1] = \begin{bmatrix} x_1[i+1] \\ x_2[i+1] \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \vec{x}[i] + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} u[i] + \vec{w}[i] \quad (3)$$

How can we treat this like two parallel problems to set this up using least-squares to get estimates for the unknown parameters $a_{11}, a_{12}, a_{21}, a_{22}, b_1, b_2$? What work/computation can we reuse across the two problems?

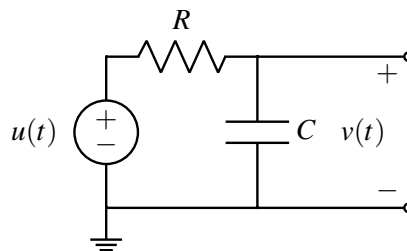
2. Discrete time system responses

We have a system $x[k+1] = \lambda x[k]$. For each λ value plotted on the real-imaginary axis, sketch $x[k]$ with an initial condition of $x[0] = 1$. Determine if each system is stable.



3. Stability Examples and Counterexamples

- (a) Consider the circuit below with $R = 1\Omega$, $C = 0.5\text{F}$, and $u(t) = \cos(t)$. Furthermore assume that $v(0) = 0$ (that the capacitor is initially discharged).



This circuit can be modeled by the differential equation

$$\frac{d}{dt}v(t) = -2v(t) + 2u(t) \quad (4)$$

Show that the differential equation is always stable. Consider what this means in the physical circuit.

- (b) Consider the discrete system

$$x[k+1] = 2x[k] + u[k] \quad (5)$$

with $x[0] = 0$.

Is the system stable or unstable? If unstable, find a bounded input sequence $u[k]$ that causes the system to “blow up”. If unstable, is there still a (non-trivial) bounded input sequence that does not cause the system to “blow up”?

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 8A

For this discussion, **Note 8** is helpful.

1. Proctoring Practice

To prepare for the midterm coming up on **March 15, 2021**, let's take a minute to ensure that the proctoring system will work.

Find the email we sent you with a Zoom link (it should have subject line "[EECS 16B] Personal Zoom Proctoring Link for Exams"), join this meeting, and record yourself for five minutes.

In case something went wrong with your Zoom room, fill out **this form** to tell us what went wrong so we can fix it for you.

2. BIBO Stability

Consider a continuous-time scalar real differential equation with known solution

$$\frac{d}{dt}x(t) = ax(t) + bu(t) \quad x(t) = e^{at}x(0) + \int_0^t e^{a(t-\tau)}bu(\tau) d\tau.$$

Show that if the system has $\operatorname{Re}\{a\} > 0$, then a bounded input can result in an unbounded output (i.e., the system is unstable) for every initial condition $x(0)$.

3. Changing behavior through feedback

In this question, we discuss how feedback control can be used to change the effective behavior of a system.

(a) Consider the scalar system:

$$x[t+1] = 0.9x[t] + u[t] + w[t] \quad (1)$$

where $u[t]$ is the control input we get to apply based on the current state and $w[t]$ is the external disturbance, each at time t .

Is the system stable? If $|w[t]| \leq \epsilon$, what can you say about $|x[t]|$ at all times t if you further assume that $u[t] = 0$ and the initial condition $x[0] = 0$? How big can $|x[t]|$ get?

(b) Suppose that we decide to choose a control law $u[t] = kx[t]$ to apply in feedback. For what values of λ can you get the system to behave like:

$$x[t+1] = \lambda x[t] + w[t]? \quad (2)$$

How would you pick k ?

(Note: In this case, $w[t]$ can be thought of like another input to the system, except we can't control it.)

(Note: In lecture we call this term f – for feedback – instead of k , but we use k here since it's a more traditional notation for feedback, and also lowercase f is confused with functions.)

(c) For the previous part, which k would you choose to minimize how big $|x[t]|$ can get?

(d) What if instead of a 0.9, we had a 3 in the original eq. (1). What, if anything, would change?

(e) Now suppose that we have a vector-valued system with a vector-valued control:

$$\vec{x}[t+1] = A\vec{x}[t] + B\vec{u}[t] + \vec{w}[t] \quad (3)$$

where we further assume that B is an invertible square matrix.

Suppose we decide to apply linear feedback control using a square matrix F so we choose $\vec{u}[t] = F\vec{x}[t]$.

For what values of matrix G can you get the system to behave like:

$$\vec{x}[t+1] = G\vec{x}[t] + \vec{w}[t] \quad (4)$$

How would you pick F given knowledge of A, B and the desired goal dynamics G ?

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 8B

The relevant note for this discussion is [Note 9](#).

1. Eigenvalues Placement in Discrete Time

Consider the following linear discrete time system

$$\vec{x}[t+1] = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} \vec{x}[t] + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u[t] + \vec{w}[t] \quad (1)$$

(a) Is the system given in eq. (1) stable?

(b) Derive a state space representation of the resulting closed loop system using state feedback of the form $u[t] = \begin{bmatrix} k_1 & k_2 \end{bmatrix} \vec{x}[t]$.

Hint: If you're having trouble parsing this expression for $u[t]$, note that $\begin{bmatrix} k_1 & k_2 \end{bmatrix}$ is a *row vector*, while $\vec{x}[t]$ is a *column vector*. What happens when we multiply a row vector with a column vector like this?

(c) Find the appropriate state feedback constants, k_1, k_2 , that place the eigenvalues of the state space representation matrix at $\lambda_1 = -\frac{1}{2}, \lambda_2 = \frac{1}{2}$.

(d) Is the system now stable?

(e) Suppose that instead of $\begin{bmatrix} 1 \\ 0 \end{bmatrix} u[t]$ in eq. (1), we had $\begin{bmatrix} 1 \\ 1 \end{bmatrix} u[t]$ as the way that the discrete-time control acted on the system. As before, we use $u[t] = \begin{bmatrix} k_1 & k_2 \end{bmatrix} \vec{x}[t]$ to try and control the system. What would the eigenvalues be? Can you move all the eigenvalues to where you want? Give an intuitive explanation of what is going on.

2. Controlling states by designing sequences of inputs

This is something that you saw in 16A in the Segway problem. In that problem, you were given a semi-realistic model for a segway. Here, we are just going to consider the following matrix chosen for ease of understanding what is going on:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Let's assume we have a *discrete-time* system defined as follows:

$$\vec{x}[t+1] = A\vec{x}[t] + \vec{b}u[t].$$

- (a) We are given the initial condition $\vec{x}[0] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$. Let's say we want to achieve $\vec{x}[T] = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ for some specific $T \geq 0$. We don't need to stay there, we just want to be in this state at that time. What is the smallest T such that this is possible? What is our choice of sequence of inputs $u[t]$?

- (b) What if we started from $\vec{x}[0] = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$? What is the smallest T and what is our choice of $u[t]$?

- (c) What if we started from $\vec{x}[0] = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}$? What is the smallest T and what is our choice of $u[t]$?

3. Uncontrollability

Consider the following discrete-time system with the given initial state:

$$\vec{x}[t+1] = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \vec{x}[t] + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u[t]$$

$$\vec{x}[0] = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

- (a) Is the system controllable?
- (b) Is it possible to reach $\vec{x}[T] = \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}$ for some $t = T$? For what input sequence $u[t]$ up to $t = T - 1$?

- (c) Is it possible to reach $\vec{x}[T] = \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix}$ for some $t = T$? For what input sequence $u[t]$ up to $t = T - 1$?
- (d) Find the set of all possible states reachable after two timesteps.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 9A

This is a review session consisting of past 16B exam problems to help you review for the midterm exam tonight.

1. Transfer Function Analysis (Fa20 MT1, Q11)

This question was presented in the form of multiple-choice and True or False; it has been adapted to mimic a format closer to this year's exam.

You are given the following transfer function:

$$H(\omega) = \frac{1}{10} \frac{\left(1 + \frac{j\omega}{100}\right) (1 + j\omega \cdot 10)}{\left(1 + \frac{j\omega}{1000}\right) (1 + j\omega)} \quad (1)$$

(a) What is the magnitude of eq. (1), $|H(\omega)|$, as $\omega \rightarrow 0$?

(b) What is $|H(\omega)|$ as $\omega \rightarrow \infty$?

(c) Now, suppose we are working with the modified transfer function below.

$$H_M(\omega) = 10 \frac{(1 + j\omega \cdot 10)}{(1 + j\omega)} \quad (2)$$

Construct the Phase Bode Plot of this transfer function. Be sure to convert the transfer function to its rational form for easy inspection of the poles and zeros. Use it to estimate the phase of the transfer function, $\angle H(\omega)$, at $\omega_1 = 1$ and $\omega_2 = 100$.

- (d) Now, compose the Bode Plot Approximation for the magnitude of the transfer function, $|H_M(\omega)|$. Use the rational form derived above.

2. CMOS Circuits (Sp20, MT1, Q1)

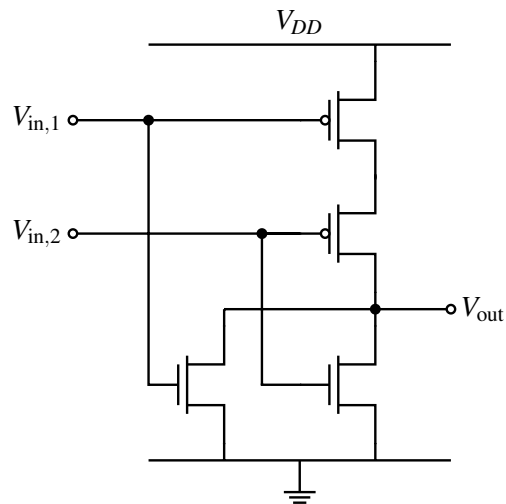


Figure 1: CMOS circuit

Consider the CMOS circuit of Figure 1. For each of the sets of $V_{in,1}$ and $V_{in,2}$ in the table below, fill in the corresponding voltage of the output V_{out} . You may assume that the threshold voltages for the transistors are $0 < V_{tn} < V_{DD}$ and $0 < |V_{tp}| < V_{DD}$.

$V_{in,1}$	$V_{in,2}$	V_o
0V	0V	
V_{DD}	0V	
0V	V_{DD}	
V_{DD}	V_{DD}	

3. Simple Differential Equation to Review!

(a) What is the solution $x(t)$ to the following scalar differential equation?

$$\frac{d}{dt}x(t) = 3jx(t) \quad x(0) = 4$$

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 9B

1. Gram-Schmidt Algorithm

Let's apply Gram-Schmidt orthonormalization to a set of three linearly independent vectors $\{\vec{s}_1, \vec{s}_2, \vec{s}_3\}$.

- (a) Find unit vector $\vec{q}_1$ such that $\text{span}(\{\vec{q}_1\}) = \text{span}(\{\vec{s}_1\})$.
- (b) Given $\vec{q}_1$ from the previous step, find $\vec{q}_2$ such that $\text{span}(\{\vec{q}_1, \vec{q}_2\}) = \text{span}(\{\vec{s}_1, \vec{s}_2\})$ and $\vec{q}_2$ is orthogonal to $\vec{q}_1$.

What would happen if $\{\vec{s}_1, \vec{s}_2, \vec{s}_3\}$ were *not* linearly independent, but rather $\vec{s}_1$ were a multiple of $\vec{s}_2$?

- (c) Now given $\vec{q}_1$ and $\vec{q}_2$ in the previous steps, find $\vec{q}_3$ such that $\text{span}(\{\vec{q}_1, \vec{q}_2, \vec{q}_3\}) = \text{span}(\{\vec{s}_1, \vec{s}_2, \vec{s}_3\})$, and $\vec{q}_3$ is orthogonal to both $\vec{q}_1$ and $\vec{q}_2$, and finally $\|\vec{q}_3\| = 1$.
- (d) Let's extend this algorithm to n linearly independent vectors. That is, given an input $\{\vec{s}_1, \dots, \vec{s}_n\}$, write the algorithm to calculate the orthonormal set of vectors $\{\vec{q}_1, \dots, \vec{q}_n\}$, where $\text{span}(\{\vec{s}_1, \dots, \vec{s}_n\}) = \text{span}(\{\vec{q}_1, \dots, \vec{q}_n\})$.

Hint: How would you calculate the i^{th} vector, $\vec{q}_i$?

2. The Order of Gram-Schmidt

If we are performing the Gram-Schmidt method on a set of vectors, does the order in which we take the vectors matter? Consider the set of vectors

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

- (a) Perform Gram-Schmidt on these vectors first in the order $\vec{v}_1, \vec{v}_2, \vec{v}_3$.
- (b) Now perform Gram-Schmidt on these vectors in the order $\vec{v}_3, \vec{v}_2, \vec{v}_1$. Do you get the same result?

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet

Discussion 10A

1. Towards Upper-Triangulation By An Orthonormal Basis

Previously in this course, we have seen the value of changing our coordinates to be eigenbasis-aligned, because we can then view the system as a cascade of scalar systems. If we have a diagonalization, then these scalar equations are fully uncoupled, and can therefore be treated completely separately. But even when we cannot diagonalize, we can upper-triangularize in a way that allows us to solve the equations one at a time, from the "bottom up".

In this problem, to better understand the steps involved, we will use the following concrete example:

$$S_{[3 \times 3]} = \begin{bmatrix} \frac{5}{12} & \frac{5}{12} & \frac{1}{6} \\ \frac{5}{12} & \frac{5}{12} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{bmatrix} \quad (1)$$

and figure out the general case by abstracting variables. Note that there is a datahub link to a jupyter notebook on the website, which will allow you to perform the numerical calculations quickly to connect the symbolic analysis to an example, but without being time-consuming in the process.¹

- (a) Consider a non-zero vector $\vec{u}_0 \in \mathbb{R}^n$. Can you think of a way to extend it to a set of basis vectors for $\mathbb{R}^n$? In other words, find $\vec{u}_1, \dots, \vec{u}_{n-1}$, such that $\text{span}(\vec{u}_0, \vec{u}_1, \dots, \vec{u}_{n-1}) = \mathbb{R}^n$. To begin with, consider $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Can you get an orthonormal basis from what you just constructed?

Hint: what was the last discussion all about? Also, the given vector isn't normalized yet!

¹This particular matrix has an additional special property of symmetry, but we won't be invoking that here.

- (b) Now consider a real eigenvalue λ_1 , and the corresponding (normalized) eigenvector $\vec{v}_1 \in \mathbb{R}^n$ of a square matrix $M \in \mathbb{R}^{n \times n}$. From the previous part, we know that we can extend $\vec{v}_1$ to an orthonormal basis of $\mathbb{R}^n$. We will denote the basis by

$$U = \begin{bmatrix} | & | & \cdots & | \\ \vec{u}_1 & \vec{u}_2 & \cdots & \vec{u}_n \\ | & | & \cdots & | \end{bmatrix}$$

where $\vec{u}_1 = \vec{v}_1$ (note that this eigenvector is already normalized).

Our goal is to look at what the matrix M looks like in the coordinate system defined by the basis U .

Compute $U^\top MU$ by writing $U = [\vec{v}_1 \quad R]$, where $R \triangleq \begin{bmatrix} | & | & \cdots & | \\ \vec{r}_1 & \vec{r}_2 & \cdots & \vec{r}_{n-1} \\ | & | & \cdots & | \end{bmatrix}$.

- (c) Show that $U^{-1} = U^\top$.

(d) Define $Q = R^\top MR$. Look at the first column and the first row of $U^\top MU$ and show that:

$$M = U \begin{bmatrix} \lambda_1 & \vec{a}^\top \\ \vec{0} & Q \end{bmatrix} U^\top$$

Here, $\vec{a}$ is a symbolic vector related to M , R , and $\vec{v}_1$ (we will show the relation!).

(e) Now, we can recurse on Q to get:

$$Q = \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix} \begin{bmatrix} \lambda_2 & \vec{b}^\top \\ \vec{0} & P \end{bmatrix} \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}^\top$$

where we have taken $\vec{v}_2 \in \mathbb{R}^{n-1}$, a normalized eigenvector of Q , associated with eigenvalue λ_2 . Again $\vec{v}_2$ is extended into an orthonormal basis to form $\begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}$.

Plug this form of Q into M above, to show that:

$$M = \begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix} \begin{bmatrix} \lambda_1 & \check{a}_1 & \check{a}_{\text{rest}}^\top \\ 0 & \lambda_2 & \vec{b}^\top \\ \vec{0} & 0 & P \end{bmatrix} \begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix}^\top$$

where we define $\check{a}$ to be the "adjusted" $\vec{a}$ to account for the substitution of Q ; $\check{a}^\top = \vec{a}^\top \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}$.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet

Discussion 10B

1. Towards upper-triangulation by an orthonormal basis

This problem is a continuation of problem 1 from Discussion 10A.

Recall that in the last discussion we set out to show that any matrix M can be upper triangularized. In particular we want to find the coordinate transformation U such that M becomes upper triangular when represented in this coordinate system:

$$T = U^{-1}MU. \quad (1)$$

In the previous discussion we began with the example of a 3×3 matrix M . We first constructed U by extending the first eigenvector of M , $\vec{v}_1$, into an orthonormal basis using Gram-Schmidt:

$$U = \begin{bmatrix} \vec{v}_1 & R \end{bmatrix}. \quad (2)$$

This gave us the transformed matrix:

$$T = \begin{bmatrix} \lambda_1 & \vec{a}^\top \\ \vec{0} & Q \end{bmatrix}, \quad (3)$$

which is upper triangular if Q is upper triangular. Then we realized that we could do a similar transformation on Q to get

$$Q = \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix} \begin{bmatrix} \lambda_2 & \vec{b}^\top \\ \vec{0} & P \end{bmatrix} \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}^\top \quad (4)$$

where $\vec{v}_2$ is the first eigenvector of matrix Q and corresponds to λ_2 . At this point, since Q was a 2×2 matrix, we can see that the transformed Q is upper triangular. We then plugged this Q in for M , and simplified to get the final result:

$$M = \begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix} \begin{bmatrix} \lambda_1 & \check{a}_1 & \check{a}_{\text{rest}} \\ 0 & \lambda_2 & \check{b} \\ 0 & 0 & P \end{bmatrix} \begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix}^\top. \quad (5)$$

- (a) Show that the matrix $\begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix}$ is still orthonormal.
- (b) We have shown how to upper triangularize a 3×3 and a 2×2 matrix. How can we generalize this process to any $n \times n$ matrix M ?
- (c) Show that the characteristic polynomial of square matrix M is the same as that of the square matrix UMU^{-1} for any invertible U .

2. Minimum Energy Control

In this question, we build up an understanding for how to get the minimum energy control signal to go from one state to another

(a) Consider the scalar system:

$$x[k+1] = ax[k] + bu[k] \quad (6)$$

where $x[0] = 0$ is the initial condition and $u[k]$ is the control input we get to apply based on the current state. Consider if we want to reach a certain state, at a certain time, namely $x[K]$. Write a matrix equation for how a choice of values of $u[k]$ for $k \in \{0, 1, \dots, K-1\}$ will determine the output at time K .

[Hint: write out all the inputs as a vector $[u[0] \ u[1] \ \dots \ u[K-2] \ u[K-1]]^T$ and figure out the combination of a and b that gives you the state at time K .]

(b) Consider the scalar system:

$$x[k+1] = 1.0x[k] + 0.7u[k] \quad (7)$$

where $x[0] = 0$ is the initial condition and $u[k]$ is the control input we get to apply based on the current state. Suppose if we want to reach a certain state, at a certain time, namely $x[K] = 14$. With our dynamics $a = 1$, solve for the best way to get to a specific state $x[K] = 14$, when $K = 10$. When we say **best way** to control a system, we want the sum squared of the inputs to be minimized

$$\operatorname{argmin}_{u[k]} \sum_{k=0}^K u[k]^2.$$

Hint: recall the Cauchy-Schwarz inequality $\langle \vec{a}, \vec{b} \rangle \leq \|\vec{a}\| \|\vec{b}\|$ where equality holds if $\vec{a}$ and $\vec{b}$ are linearly dependent.

(c) Consider the scalar system:

$$x[t+1] = 0.5x[t] + 0.7u[t] \quad (8)$$

where $x[0] = 0$ is the initial condition and $u[t]$ is the control input we get to apply based on the current state. Consider if we want to reach a certain state, at a certain time, namely $x[K] = 14$, when $K = 10$. Explain in words the trend of the control input that will be used to solve this problem.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 11A

In this discussion, we discuss orthonormal transformations in the context of setting up a least squares problem, and in the end, we make a connection to the SVD.

1. Orthonormality, Least Squares, and Intro to SVD

- (a) Let U be an $m \times n$ matrix with orthonormal columns, with $m \geq n$. Compute $U^\top U$. How does this change if $m < n$?

- (b) Suppose you have a real, square, $n \times n$ orthonormal matrix U (the columns of U are unit norm and mutually orthogonal). You also have real vectors $\vec{x}_1, \vec{x}_2, \vec{y}_1, \vec{y}_2$ such that

$$\begin{aligned}\vec{y}_1 &= U\vec{x}_1 \\ \vec{y}_2 &= U\vec{x}_2\end{aligned}$$

Calculate $\langle \vec{y}_1, \vec{y}_2 \rangle = \vec{y}_2^\top \vec{y}_1 = \vec{y}_1^\top \vec{y}_2$ in terms of $\langle \vec{x}_1, \vec{x}_2 \rangle = \vec{x}_2^\top \vec{x}_1 = \vec{x}_1^\top \vec{x}_2$.

(c) Following the previous question, express $\|\vec{y}_1\|_2^2$ and $\|\vec{y}_2\|_2^2$ in terms of $\|\vec{x}_1\|_2^2$ and $\|\vec{x}_2\|_2^2$.

(d) Suppose you observe data coming from the model $y_i = \vec{a}^\top \vec{x}_i$, and you want to find the linear scale-parameters (each a_i). We are trying to learn the model $\vec{a}$. You have m data points $(\vec{x}_i, y_i)$, with each $\vec{x}_i \in \mathbb{R}^n$.

Note that $\vec{x}_i$ refers to the i -th vector, not the i -th element of a single vector. Each $\vec{x}_i$ is a different input vector that you take the inner product of with $\vec{a}$, giving a scalar y_i .

Set up a least squares formulation for estimating $\vec{a}$, and find the solution to the least squares problem.

(e) Now suppose V is an orthonormal square matrix, and rather than observing $\vec{a}^\top \vec{x}$ directly, we actually observe data points that result from our inputs being transformed by $V^\top$ as follows:

$$\tilde{\vec{x}} = V^\top \vec{x} \quad (1)$$

That is, our model acts on the modified input data $\tilde{\vec{x}}_i$, so the data points we collected are now $(\tilde{\vec{x}}, y)$.

We must now consider the new model:

$$y = \tilde{\vec{a}}^\top \tilde{\vec{x}} \quad (2)$$

$$= \tilde{\vec{a}}^\top V^\top \vec{x} \quad (3)$$

Set up a least-squares formulation for $\hat{\vec{a}}$. How is $\hat{\vec{a}}$ related to $\hat{\vec{a}}$?

(f) Now suppose that we have the matrix

$$\begin{bmatrix} \vec{x}_1^\top \\ \vec{x}_2^\top \\ \vdots \\ \vec{x}_m^\top \end{bmatrix} \triangleq X = U \Sigma V^\top. \quad (4)$$

where U is an $m \times m$ matrix, and V is an $n \times n$ matrix. Suppose that we have more data points than the dimension of our space (that is, $m > n$). Also, the transformation V in part e) is the same V in this full SVD representation. Set up a least squares formulation for estimating $\vec{a}$ and find the solution to the least squares. Is there anything interesting going on?

Note: Don't worry about how we would find u , Σ , $V^\top$ for now; assume that X has the given form and that U and V are orthonormal.

Hint: Start by substituting the full SVD representation of X into the answer of the previous part.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 11B

1. Understanding the SVD

We can compute the SVD for a wide matrix A with dimension $m \times n$ where $n > m$ using $A^\top A$ with the method covered in lecture. However, when doing so, you may realize that $A^\top A$ is much larger than $AA^\top$ for such wide matrices. This makes it more efficient to find the eigenvalues for $AA^\top$. In this question, we will explore how to compute the SVD using $AA^\top$ instead of $A^\top A$.

(a) What are the dimensions of $AA^\top$ and $A^\top A$?

(b) Given that the SVD of A is $A = U\Sigma V^\top$, find a symbolic expression for $AA^\top$.

(c) Using the solution to the previous part, how can we find U and Σ from $AA^\top$?

(d) Now that we have found the singular values σ_i and the corresponding vectors $\vec{u}_i$ in the matrix U , can you find the corresponding vectors $\vec{v}_i$ in V ?

(e) Now we have a way to find the vectors $\vec{v}_i$ in matrix V ! Verify that these vectors are orthonormal.

(f) Now that we have found $\vec{v}_i$, you may notice that we only have $m < n$ vectors of dimension n . This is not enough for a basis. How would you complete the m vectors to form an orthonormal basis?

(g) (Practice.) Given that $A = U\Sigma V^\top$ verify that the vectors you found to extend the $\vec{v}_i$ into a basis are in the nullspace of A .

(h) Using the previous parts of this question and what you learned from lecture, write out a procedure on how to find the SVD for *any* matrix.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 12A

In this discussion, we practice computing the SVD for a "wide" matrix (more columns than rows) and for a "tall" matrix (more rows than columns). There is also an associated jupyter notebook on Datahub that will serve useful to confirm the numerical calculations (specifically for performing Gram-Schmidt).

Also, note that the techniques and insights communicated in this discussion are conveyed in [Note 13, sec. 3](#).

1. Computing the SVD: A "Tall" Matrix Example

Define the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}.$$

(a) In this part, we will find the full SVD of A in steps.

(i) Compute $A^T A$ and find its eigenvalues.

(ii) Find orthonormal eigenvectors $\vec{v}_i$ (right singular vectors, columns of V).

(iii) Find singular values, $\sigma_i = \sqrt{\lambda_i}$.

(iv) Use $\vec{v}_i$ to find orthonormal $\vec{u}_i$ (for nonzero σ).

(v) Use the previous parts to write the full SVD of A .

(vi) Use the Jupyter notebook to run the code cell that calls `numpy.linalg.svd` on A . What is the result? Does it match our result above?

(b) Find the rank of A .

(c) Find a basis for the range (or column space) of A .

(d) Find a basis for the null space of A .

(e) We now want to create the SVD of $A^\top$. Rather than repeating all of the steps in the algorithm, feel free to use the jupyter notebook for this subpart (which defines a `numpy.linalg.svd` command). What are the relationships between the matrices composing A and the matrices composing $A^\top$?

2. Computing the SVD: A "Wide" Matrix Example

Define the matrix

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}.$$

- (a) In this part, we will find the full SVD of A in steps.
 - (i) Compute AA^T and find its eigenvalues.
 - (ii) Find orthonormal eigenvectors $\vec{u}_i$ (left singular vectors, columns of U). Feel free to use the associated Jupyter notebook to perform Gram-Schmidt for this part, if needed.
 - (iii) Find the singular values, $\sigma_i = \sqrt{\lambda_i}$.
 - (iv) Use $\vec{u}_i$ to find orthonormal $\vec{v}_i$ (for nonzero σ). Feel free to use the associated Jupyter notebook to perform Gram-Schmidt for this part, if needed.

- (v) Use the previous parts to write the full SVD of A .

- (b) Find the rank of A , using the computed full SVD.

- (c) Find a basis for the range (or columnspace) of A .

- (d) Find a basis for the nullspace of A .

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 12B

1. Geometric interpretation of the SVD

In this exercise, we explore the geometric interpretation of symmetric matrices and how this connects to the SVD. We consider how a real 2×2 matrix acts on the unit circle, transforming it into an ellipse. It turns out that the principal semiaxes of the resulting ellipse are related to the singular values of the matrix, as well as the vectors in the SVD.

(a) Consider the real 2×2 matrix

$$A = \begin{bmatrix} 0 & -1 \\ 3 & 0 \end{bmatrix}. \quad (1)$$

Also consider the unit circle in $\mathbb{R}^2$,

$$S = \left\{ \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \middle| 0 \leq \theta < 2\pi \right\}. \quad (2)$$

Plot the transformed circle, AS , on the $\mathbb{R}^2$ plane.

(b) Now let's consider how this transformation looks in the lens of the SVD. The SVD for matrix A is:

$$A = U\Sigma V^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad (3)$$

$$A\vec{x} = U\Sigma V^T \vec{x} = U \left(\Sigma \left(V^T \vec{x} \right) \right). \quad (4)$$

Let's start by examining the effects of each of these matrices one at a time, right to left, in the same order that they would be applied to a vector $\vec{x}$.

What does the unit circle look like after being transformed by just V^T ? Plot $S_1 = V^T S$ on the $\mathbb{R}^2$ plane. Geometrically speaking, what does V^T do to any given $\vec{x}$.

(c) What does the unit circle look like after being transformed by ΣV^T ? Plot $S_2 = \Sigma V^T S$ on the $\mathbb{R}^2$ plane. Geometrically speaking, what is the Σ matrix doing to any given $V^T \vec{x}$?

(d) What does the unit circle look like after being transformed by $U\Sigma V^\top$? Plot $S_3 = U\Sigma V^\top S$ on the $\mathbb{R}^2$ plane. Geometrically speaking, what is the U matrix doing to any given $\Sigma V^\top \vec{x}$?

(e) Consider the columns of the matrices U, V from the SVD of A in part (b), and treat them as vectors in $\mathbb{R}^2$. Let $U = (\vec{u}_1 \ \vec{u}_2)$, $V = (\vec{v}_1 \ \vec{v}_2)$. Let σ_1, σ_2 be the singular values of A , where $\sigma_1 \geq \sigma_2$. In your plot of AS , draw the vectors $\sigma_1 \vec{u}_1$ and $\sigma_2 \vec{u}_2$ from the origin. What do these vectors correspond to geometrically?

(f) Repeat parts (b-e) for the following matrices, and note down any interesting things you notice.

i. A rotation matrix, $A_1 = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$.

ii. A diagonal matrix, $A_2 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$.

iii. A symmetric matrix, $A_3 = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$.

iv. A matrix with non-trivial nullspace, $A_4 = \begin{bmatrix} 4 & 2 \\ -2 & -1 \end{bmatrix}$.

v. An arbitrary matrix, $A_5 = \begin{bmatrix} 1.6 & 2.4 \\ -0.4 & -1 \end{bmatrix}$.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 13A

This discussion will recap a lot of the key concepts covered in **lecture last week**.

1. Linear Approximation

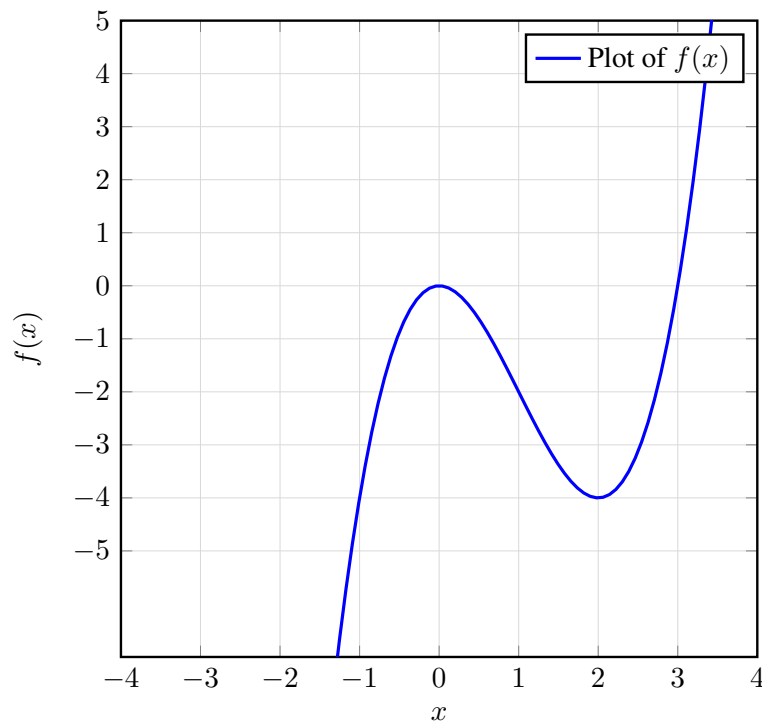
A common way to approximate a nonlinear function is to perform linearization near a point. In the case of a one-dimensional function $f(x)$, the linear approximation of $f(x)$ at a point x_* is given by

$$f(x) \approx f(x_*) + f'(x_*) \cdot (x - x_*), \quad (1)$$

where $f'(x_*) := \left. \frac{df(x)}{dx} \right|_{x=x_*}$ is the derivative of $f(x)$ at $x = x_*$.

Keep in mind that wherever we see x_* , this denotes a *constant value* or operating point.

- (a) Suppose we have the single-variable function $f(x) = x^3 - 3x^2$. We can plot the function $f(x)$ as follows:



- i. Write the linear approximation of the function around an arbitrary point x_* .

- ii. Use the expression above to linearize the function around the point $x = 1.5$. Draw the linearization into the plot of part i).

Now that we have this specific point's linearization, we understand how the function behaves around the point. Let's use this linearization to evaluate the function's approximation at $x = 1.7$ (based on our approximation at $x = 1.5$, we want to see how a $\delta = +0.2$ shift in the x value changes the corresponding $f(x)$ value). How does this approximation compare to the exact value of the function at $x = 1.7$?

$$f(1.7) \approx -3.375 + (-2.25) \cdot (1.7 - 1.5) \quad (2)$$

$$\approx -3.375 - 0.45 \quad (3)$$

$$\approx -3.825 \quad (4)$$

Comparing to the exact value $f(1.7) = 1.7^3 - 3 \cdot 1.7^2 = -3.757$, we find that the difference is 0.068. Not too bad! What if we repeat with $\delta = 1$? To do so, we must use the approximation around $x = 1.5$ to compute $x = 2.5$, and compare to the exact value $f(2.5)$. How does our new approximation compare to the exact result?

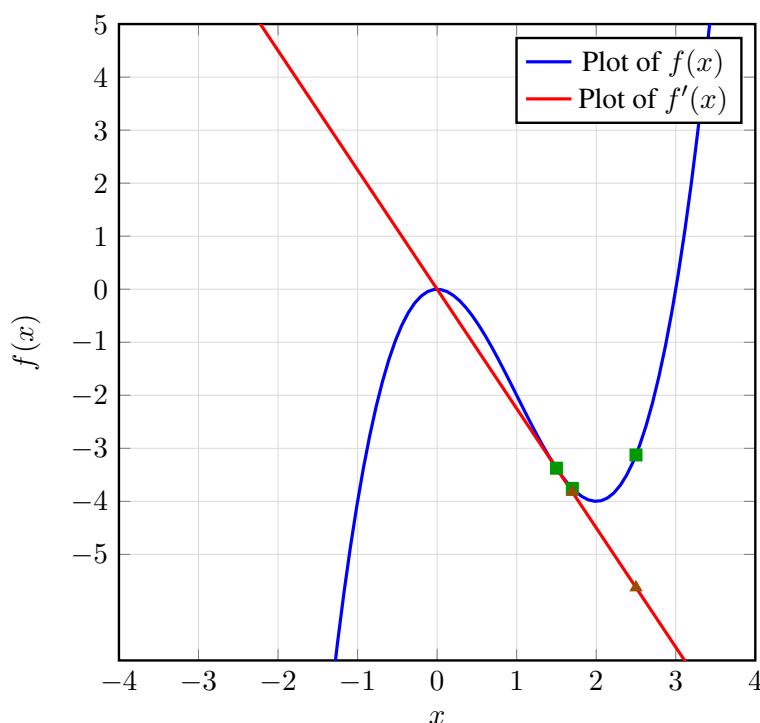
$$f(2.5) \approx -3.375 + (-2.25) \cdot (2.5 - 1.5) \quad (5)$$

$$\approx -3.375 - 2.25 \quad (6)$$

$$\approx -5.625 \quad (7)$$

Comparing to the exact value $f(2.5) = 2.5^3 - 3 \cdot 2.5^2 = -3.125$, we find that the difference is much larger; the error jumped to 2.5! This is an error multiplication of $\frac{2.5}{0.068} \approx 37$, even though our δ only multiplied by 5. What happened?

Looking at the actual function, we see that the function has a significant curvature between our "anchor point" of $x_{\star} = 1.5$ and $x = 2.5$. Our linear model is unable to capture this curvature, and so we estimated $f(2.5)$ as if the function kept decreasing, as it did around $x = 1.5$ (where the slope was -2.25).



Now, we can extend this to higher dimensional functions. In the case of a two-dimensional function $f(x, y)$, the linear approximation of $f(x, y)$ at a point (x_*, y_*) is given by

$$f(x, y) \approx f(x_*, y_*) + f_x(x_*, y_*) \cdot (x - x_*) + f_y(x_*, y_*) \cdot (y - y_*). \quad (8)$$

where $f_x(x_*, y_*)$ is the partial derivative of $f(x, y)$ with respect to x at the point (x_*, y_*) :

$$f_x(x_*, y_*) = \left. \frac{\partial f(x, y)}{\partial x} \right|_{(x_*, y_*)} \quad (9)$$

and $f_y(x_*, y_*)$ is the partial derivative of $f(x, y)$ with respect to y at the point (x_*, y_*) .

- (b) Now, let's see how we can derive partial derivatives. When we are given a function $f(x, y)$, we calculate the partial derivative of f with respect to x by fixing y and taking the derivative with respect to x . Given the function $f(x, y) = x^2y$, find the partial derivatives $f_y(x, y)$ and $f_x(x, y)$.

- (c) Write out the linear approximation of f near (x_*, y_*) .

- (d) We want to see if the approximation arising from linearization of this function is reasonable for a point close to our point of evaluation. First, approximate $f(x, y)$ at the point $(2.01, 3.01)$ using $(x_*, y_*) = (2, 3)$, and compare the result to $f(2.01, 3.01)$.

- (e) Suppose we have now a vector-valued function $f(\vec{x}, \vec{y})$, which takes in vectors $\vec{x} \in \mathbb{R}^n$, $\vec{y} \in \mathbb{R}^k$ and outputs a scalar $\in \mathbb{R}$. That is, $f(\vec{x}, \vec{y})$ is $\mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}$. With this new model, how can we adapt our previous linearization method?

One way to linearize the function f is to do it for every single element in $\vec{x} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}^\top$ and $\vec{y} = \begin{bmatrix} y_1 & y_2 & \dots & y_k \end{bmatrix}^\top$. Then, when we are looking at x_i or y_j , we fix everything else as constant. This would give us the linear approximation

$$f(\vec{x}, \vec{y}) \approx f(\vec{x}_*, \vec{y}_*) + \sum_{i=1}^n \frac{\partial f(\vec{x}, \vec{y})}{\partial x_i} (x_i - x_{*,i}) + \sum_{j=1}^k \frac{\partial f(\vec{x}, \vec{y})}{\partial y_j} (y_j - y_{*,j}). \quad (10)$$

In order to simplify this equation, we can define the rows $D_{\vec{x}}$ and $D_{\vec{y}}$ as

$$D_{\vec{x}}f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix}, \quad (11)$$

$$D_{\vec{y}}f = \begin{bmatrix} \frac{\partial f}{\partial y_1} & \dots & \frac{\partial f}{\partial y_k} \end{bmatrix}. \quad (12)$$

Then, Equation (10) can be rewritten as

$$f(\vec{x}, \vec{y}) \approx f(\vec{x}_*, \vec{y}_*) + (D_{\vec{x}}f) \Big|_{(\vec{x}_*, \vec{y}_*)} \cdot (\vec{x} - \vec{x}_*) + (D_{\vec{y}}f) \Big|_{(\vec{x}_*, \vec{y}_*)} \cdot (\vec{y} - \vec{y}_*). \quad (13)$$

Assume that $n = k$ and we define the function $f(\vec{x}, \vec{y}) = \vec{x}^\top \vec{y} = \sum_{i=1}^k x_i y_i$. Find $D_{\vec{x}}f$ and $D_{\vec{y}}f$.

- (f) Following the above part, find the linear approximation of $f(\vec{x}, \vec{y})$ near $\vec{x}_\star = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{y}_\star = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

Recall that $f(\vec{x}, \vec{y}) = \vec{x}^\top \vec{y} = \sum_{i=1}^k x_i y_i$.

- (g) When the function $\vec{f}(\vec{x}, \vec{y}) : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^m$ takes in vectors and outputs a vector (rather than a scalar), we can view each dimension in $\vec{f}$ independently as a separate function f_i , and linearize each of them:

$$\vec{f}(\vec{x}, \vec{y}) = \begin{bmatrix} f_1(\vec{x}, \vec{y}) \\ f_2(\vec{x}, \vec{y}) \\ \vdots \\ f_m(\vec{x}, \vec{y}) \end{bmatrix} \approx \begin{bmatrix} f_1(\vec{x}_*, \vec{y}_*) + D_{\vec{x}}f_1 \cdot (\vec{x} - \vec{x}_*) + D_{\vec{y}}f_1 \cdot (\vec{y} - \vec{y}_*) \\ f_2(\vec{x}_*, \vec{y}_*) + D_{\vec{x}}f_2 \cdot (\vec{x} - \vec{x}_*) + D_{\vec{y}}f_2 \cdot (\vec{y} - \vec{y}_*) \\ \vdots \\ f_m(\vec{x}_*, \vec{y}_*) + D_{\vec{x}}f_m \cdot (\vec{x} - \vec{x}_*) + D_{\vec{y}}f_m \cdot (\vec{y} - \vec{y}_*) \end{bmatrix} \quad (14)$$

We can rewrite this in a clean way with the *Jacobian*:

$$D_{\vec{x}}\vec{f} = \begin{bmatrix} D_{\vec{x}}f_1 \\ D_{\vec{x}}f_2 \\ \vdots \\ D_{\vec{x}}f_m \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}, \quad (15)$$

and similarly

$$D_{\vec{y}}\vec{f} = \begin{bmatrix} \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_k} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial y_1} & \cdots & \frac{\partial f_m}{\partial y_k} \end{bmatrix}. \quad (16)$$

Then, the linearization becomes

$$\vec{f}(\vec{x}, \vec{y}) \approx \vec{f}(\vec{x}_*, \vec{y}_*) + (D_{\vec{x}}\vec{f})\Big|_{(\vec{x}_*, \vec{y}_*)} \cdot (\vec{x} - \vec{x}_*) + (D_{\vec{y}}\vec{f})\Big|_{(\vec{x}_*, \vec{y}_*)} \cdot (\vec{y} - \vec{y}_*). \quad (17)$$

Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\vec{f}(\vec{x}) = \begin{bmatrix} x_1^2 x_2 \\ x_1 x_2^2 \end{bmatrix}$. Find $D_{\vec{x}}\vec{f}$, applying the definition above.

- (h) Compare the approximation of $\vec{f}$ at the point $\begin{bmatrix} 2.01 \\ 3.01 \end{bmatrix}$ using $\vec{x}_\star = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ versus $\vec{f}\left(\begin{bmatrix} 2.01 \\ 3.01 \end{bmatrix}\right)$. Recall the definition that $\vec{f}(\vec{x}) = \begin{bmatrix} x_1^2 x_2 \\ x_1 x_2^2 \end{bmatrix}$.

- (i) **Practice:** Let $\vec{x}$ and $\vec{y}$ be vectors with 2 rows, and let $\vec{w}$ be another vector with 2 rows. Let $\vec{f}(\vec{x}, \vec{y}) = \vec{x}\vec{y}^\top \vec{w}$. Find $D_{\vec{x}}\vec{f}$ and $D_{\vec{y}}\vec{f}$.

(j) **Practice:** Continuing the above part, find the linear approximation of $\vec{f}$ near $\vec{x} = \vec{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and with

$$\vec{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

These linearizations are important for us because we can do many easy computations using linear functions.

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet

Discussion 13B

1. Quadratic Approximation and Vector Differentiation

As shown in the previous discussion, a common way to approximate a non-linear high-dimensional functions is to perform linearization near a point. In the case of a two-dimensional function $f(x, y)$ with scalar output, the linear approximation of $f(x, y)$ at a point (x_*, y_*) is given by

$$f(x, y) \approx f(x_*, y_*) + f_x(x_*, y_*)(x - x_*) + f_y(x_*, y_*)(y - y_*) \quad (1)$$

where as in the previous section,

$$f_x(x_*, y_*) = \left. \frac{\partial f(x, y)}{\partial x} \right|_{(x_*, y_*)} \quad \text{and} \quad f_y(x_*, y_*) = \left. \frac{\partial f(x, y)}{\partial y} \right|_{(x_*, y_*)}. \quad (2)$$

In vector form, this can be written as:

$$f(\vec{x}) \approx f(\vec{x}_*) + \left[D_{\vec{x}} f|_{\vec{x}_*} \right] (\vec{x} - \vec{x}_*). \quad (3)$$

Recall from the previous discussion that $D_{\vec{x}} f$ is a row-vector filled with the partial derivatives $\frac{\partial f(\vec{x})}{\partial x_i}$:

$$D_{\vec{x}} f = \left[\frac{\partial f(\vec{x})}{\partial x_1} \quad \dots \quad \frac{\partial f(\vec{x})}{\partial x_n} \right] = \left[f_{x_1}(\vec{x}) \quad \dots \quad f_{x_n}(\vec{x}) \right]. \quad (4)$$

Our goal is to extend this idea to a quadratic approximation. To do this, we need some notion of a second derivative.

For this discussion, we will only be considering these types of functions from $\mathbb{R}^n \rightarrow \mathbb{R}$, since that is the typical form for a cost function used during optimization.

- (a) Given the function $f(x) = e^{-2x}$, find the first and second derivatives, and write out its quadratic approximation at $x = x_*$.

Hint: Use Taylor's theorem.

- (b) To write second partial derivatives compactly, we will introduce a new notation that builds off the notation f_x and f_y introduced previously. To compute f_{xy} , we first take the derivative in x , then in y :

$$f_{xy}(x_*, y_*) = \left. \frac{\partial f_x(x, y)}{\partial y} \right|_{(x_*, y_*)} = \left. \frac{\partial^2 f(x, y)}{\partial y \partial x} \right|_{(x_*, y_*)}. \quad (5)$$

Given the function $f(x, y) = x^2 y^2$, find all of the first and second partial derivatives.

- (c) To find the quadratic approximation of $f(x, y)$ near (x_*, y_*) , we plug in $f(x_* + \Delta x, y_* + \Delta y)$ and drop the terms that are higher order than quadratic:

$$f(x_* + \Delta x, y_* + \Delta y) = (x_* + \Delta x)^2 (y_* + \Delta y)^2 \quad (6)$$

$$= (x_*^2 + 2x_*\Delta x + (\Delta x)^2)(y_*^2 + 2y_*\Delta y + (\Delta y)^2) \quad (7)$$

$$\approx x_*^2 y_*^2 + 2x_* y_*^2 \Delta x + 2x_*^2 y_* \Delta y \quad (8)$$

$$+ y_*^2 (\Delta x)^2 + 4x_* y_* (\Delta x)(\Delta y) + x_*^2 (\Delta y)^2 \quad (9)$$

$$= f(x_*, y_*) + f_x(x_*, y_*)\Delta x + f_y(x_*, y_*)\Delta y \quad (10)$$

$$+ \frac{1}{2} f_{xx}(x_*, y_*) (\Delta x)^2 + \frac{1}{2} f_{yy}(x_*, y_*) (\Delta y)^2 \quad (11)$$

$$+ f_{xy}(x_*, y_*) (\Delta x)(\Delta y). \quad (12)$$

This is slightly different from the expression we get via the Taylor series expansion. How would we rewrite this expression, so that *all* second derivatives are involved, each with a coefficient of $\frac{1}{2}$?

- (d) Just as we created the derivative row vector to hold all the first partial derivatives to help in writing linearization in matrix/vector form:

$$D_{\vec{x}}f = \left[\frac{\partial f(\vec{x})}{\partial x_1} \quad \cdots \quad \frac{\partial f(\vec{x})}{\partial x_n} \right] = \left[f_{x_1}(\vec{x}) \quad \cdots \quad f_{x_n}(\vec{x}) \right] \quad (13)$$

we would like to create a matrix to hold all the second partial derivatives to help in writing quadratic approximation in matrix/vector form:

$$H_{\vec{x}}f = \begin{bmatrix} \frac{\partial^2 f(\vec{x})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\vec{x})}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\vec{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\vec{x})}{\partial x_n^2} \end{bmatrix} = \begin{bmatrix} f_{x_1 x_1}(\vec{x}) & \cdots & f_{x_n x_1}(\vec{x}) \\ \vdots & \ddots & \vdots \\ f_{x_1 x_n}(\vec{x}) & \cdots & f_{x_n x_n}(\vec{x}) \end{bmatrix} \quad (14)$$

This matrix is the *Hessian* of f . Note that this quantity is different from the *Jacobian* matrix that was covered in the previous discussion. In contrast to the Hessian, which is the matrix of second partial derivatives of a *scalar-valued vector-input* function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the Jacobian is the matrix of first partial derivatives of a *vector-valued vector-input* function $\vec{f}: \mathbb{R}^n \rightarrow \mathbb{R}^k$.

In fact, the Hessian is the (Jacobian) derivative of the derivative; if we let $\vec{g}(\vec{x}) = (D_{\vec{x}}f)^\top$ (so that it's a column vector and the dimensions work out), then $H_{\vec{x}}f = D_{\vec{x}}\vec{g}$.

To get a feel for the Hessian of f , find $H_{(x,y)}f$ for the f above, that is, $f(x, y) = x^2 y^2$.

- (e) Using the Hessian, write out the general formula for the quadratic approximation of a scalar-valued function f of a vector $\vec{x}$ in vector/matrix form.

- (f) Show that the quadratic approximation for the scalar-valued function $f(\vec{w}) = e^{\vec{x}^\top \vec{w}}$ around $\vec{w} = \vec{w}_*$ is

$$f(\vec{w}_* + \Delta \vec{w}) \approx e^{\vec{x}^\top \vec{w}_*} \left(1 + \vec{x}^\top (\Delta \vec{w}) + \frac{1}{2} \left(\vec{x}^\top (\Delta \vec{w}) \right)^2 \right). \quad (15)$$

Here, assume that $\vec{x}$ is just some given vector — a constant vector.

Hint: You can compute the following partial derivatives:

$$f_{w_i}(\vec{w}) = x_i f(\vec{w}) \quad (16)$$

$$f_{w_i w_j}(\vec{w}) = x_i x_j f(\vec{w}). \quad (17)$$

Now compute $D_{\vec{w}} f$ and $H_{\vec{w}} f$, and plug it into the quadratic approximation formula.

- (g) Use linearity to give the quadratic approximation for the function $\sum_{i=1}^m e^{\vec{x}_i^\top \vec{w}}$ around $\vec{w} = \vec{w}_*$. Here, assume that the $\vec{x}_i$ are just some given vectors.

- (h) **Practice.** The second derivative also has an interpretation as the derivative of the derivative. However, we saw that the derivative of a scalar-valued function with respect to a vector is naturally a row. If you

wanted to approximate how much the derivative changed by moving a small amount $\Delta \vec{w}$, how would you get such an estimate using your expression for the second derivative?

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 14A

1. Complex Inner Products, Projections, and Orthonormality

To understand how we want to define complex inner products, it is useful to first recall how we came to define real inner products. Inner products grow out of our desire to do projections. Projections themselves are intimately connected to the idea of orthogonality.

When projecting a vector $\vec{v}$ onto another vector $\vec{u}$, the result needs to be $r\vec{u}$ where r is some constant. In other words, we want a vector that is linearly dependent with $\vec{u}$ so that it captures all of $\vec{v}$ that is in the direction of $\vec{u}$.

Because the idea of *direction* is so important, we can first focus on the distilled embodiments of directions themselves — namely unit vectors. Vectors whose length is 1 essentially are all about direction since their magnitude/length is known.

If we have a vector $\vec{u} \in \mathbb{R}^n$, recall that we can define the projection operator as $P_{\vec{u}} = \frac{\vec{u}\vec{u}^\top}{\|\vec{u}\|^2}$ that acts on vectors. This means, for any vector $\vec{v}$, we can project $\vec{v}$ onto the vector $\vec{u}$ by computing the projection $P_{\vec{u}}\vec{v}$ ¹:

$$P_{\vec{u}}\vec{v} = \frac{\vec{u}\vec{u}^\top}{\|\vec{u}\|^2}\vec{v} = \frac{\vec{u}^\top\vec{v}}{\|\vec{u}\|^2}\vec{u}. \quad (1)$$

This gives us a vector that is in the direction of $\vec{u}$ and is a multiple $\vec{u}^\top\vec{v}$ of $\vec{u}$. This is a projection because the residual $\vec{v} - P_{\vec{u}}\vec{v}$ is orthogonal to $\vec{u}$:

$$\vec{u}^\top(\vec{v} - P_{\vec{u}}\vec{v}) = \vec{u}^\top\left(\vec{v} - \frac{\vec{u}^\top\vec{v}}{\|\vec{u}\|^2}\vec{u}\right) = \vec{u}^\top\vec{v} - \frac{(\vec{u}^\top\vec{u})(\vec{u}^\top\vec{v})}{\|\vec{u}\|^2} = \vec{u}^\top\vec{v} - \vec{u}^\top\vec{v} = 0. \quad (2)$$

- (a) First, we want to consider formulating the projection with real inner products instead of norms. Recall that we defined the real inner product as $\langle \vec{u}, \vec{v} \rangle = \sum_i u_i v_i = \vec{u}^\top \vec{v} = \vec{v}^\top \vec{u}$. Use this to rewrite the matrix $P_{\vec{u}}$ just using inner products instead of norms.

Use this to show that when we project a vector $\vec{v}$ onto a vector $\vec{u}$, that the result is $\frac{\langle \vec{v}, \vec{u} \rangle}{\langle \vec{u}, \vec{u} \rangle} \vec{u}$.

¹For example, in the case of least squares, this projection would be onto $\text{col}(A)$ and be of the form $P_{\text{col}(A)}(\vec{b}) = A\vec{x} = A(A^\top A)^{-1}A^\top \vec{b}$.

- (b) Now, let's expand this to the case where the vectors are complex.

To be precise, an n -dimensional *complex vector* is just like an n -dimensional vector that we're used to working with, *but* each of the components are complex numbers instead of real numbers.

One-dimensional real vectors are just 1×1 vectors of the form $\begin{bmatrix} r \end{bmatrix}$, for $r \in \mathbb{R}$. Correspondingly, one-dimensional complex vectors are just vectors of the form $\begin{bmatrix} c \end{bmatrix}$, for $c \in \mathbb{C}$. Make sure to keep in mind the difference between these vectors, and the familiar real/complex numbers.

If $\vec{v}$ is a complex vector, we define its length, or norm, by the equality

$$\|\vec{v}\|^2 = \sum_{i=1}^n |v_i|^2 = \sum_{i=1}^n v_i \overline{v_i}. \quad (3)$$

Here, recall that $\overline{v_i}$ is the complex conjugate of the complex number v_i . Notice that this is similar *but not exactly equal* to the norm on real vectors defined by $\|\vec{v}\|^2 = \sum_{i=1}^n v_i^2$; in particular, $v_i \overline{v_i} \neq v_i^2$, unless v_i is real.

A *complex unit vector* is a complex vector with norm 1, exactly analogous to real vectors with norm 1 being the “regular” unit vectors we're used to working with.

The one-dimensional real unit vectors are just $\begin{bmatrix} 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \end{bmatrix}$, since $|1|^2 = |-1|^2 = 1$. What are the one-dimensional complex unit vectors?

- (c) If we are considering *real* vectors, then two vectors $\vec{v}_1$ and $\vec{v}_2$ are *linearly dependent* if $\vec{v}_1 \in \text{span}(\vec{v}_2)$. That is, there is a real scalar $r \in \mathbb{R}$ such that $\vec{v}_1 = r\vec{v}_2$. In this case, r is the coefficient of the projection of $\vec{v}_1$ onto $\vec{v}_2$, and in particular $r = \frac{\langle \vec{v}_1, \vec{v}_2 \rangle}{\langle \vec{v}_2, \vec{v}_2 \rangle}$.

Now let's consider the case of *complex* vectors. Two complex vectors $\vec{v}_1$ and $\vec{v}_2$ are *linearly dependent* if there is a complex scalar $c \in \mathbb{C}$ such that $\vec{v}_1 = c\vec{v}_2$. In this way, even though we haven't defined a way to project general complex vectors onto other complex vectors yet, we already know the projection coefficient c ² of $\vec{v}_1$ onto $\vec{v}_2$. (Remember, the only reason we can do this is because they're linearly dependent, a very special case).

The previous problem had us pick out that the one-dimensional complex unit vectors are $\begin{bmatrix} e^{j\theta} \end{bmatrix}$ for $0 \leq \theta < 2\pi$. Two special cases are $\begin{bmatrix} 1 \end{bmatrix} = \begin{bmatrix} e^{j \cdot 0} \end{bmatrix}$ and $\begin{bmatrix} j \end{bmatrix} = \begin{bmatrix} e^{j \cdot \frac{\pi}{2}} \end{bmatrix}$. Calculate the projection coefficient of the projection of $\begin{bmatrix} 1 \end{bmatrix}$ onto $\begin{bmatrix} j \end{bmatrix}$ and for the projection of $\begin{bmatrix} j \end{bmatrix}$ onto $\begin{bmatrix} 1 \end{bmatrix}$.

²Remember, this is not the projection! This coefficient is specifically a scalar value – projections are vector-valued.

- (d) The previous example has shown you that when complex numbers are involved, the order matters of who is being projected onto whom, even when both are unit vectors. Now that we do not have symmetry, we need to be more careful in formulating a projection operator for complex vectors.

Consider a complex vector $\vec{u}$. Define the following operator: $P_{\vec{u}} = \frac{\vec{u}\vec{u}^*}{\|\vec{u}\|^2}$. Here $\vec{u}^* = \left(\overline{\vec{u}}\right)^\top$ is the conjugate transpose of the vector $\vec{u}$ – in other words, we take the complex conjugate of every entry in $\vec{u}$, and take the transpose of the result. For real vectors $\vec{u}$, this is the same as the projection operator we had above because the complex conjugate would do nothing.

We propose that it's the projection operator in the complex case, that is, $P_{\vec{u}}\vec{v}$ is the projection of $\vec{v}$ onto the span of $\vec{u}$. We don't know for sure this is the case – we haven't proved it, only guessed it, after all – but we plan to show it a little later, once we do an example.

In the 1-d case, suppose $\vec{u} = \begin{bmatrix} c \end{bmatrix}$ is a complex unit vector. Then

$$P_{\vec{u}} = \frac{\vec{u}\vec{u}^*}{\|\vec{u}\|^2} = \frac{1}{|c|^2} \begin{bmatrix} c \end{bmatrix} \begin{bmatrix} \bar{c} \end{bmatrix} = \frac{1}{|c|^2} \begin{bmatrix} c\bar{c} \end{bmatrix} = \frac{1}{|c|^2} \begin{bmatrix} |c|^2 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} = I_1. \quad (4)$$

Now let's check the 2-d case. As in the 1-d case, we want our vectors to be linearly dependent at first, to check our intuition. For two unit vectors $\vec{v}_1 = \begin{bmatrix} \frac{j}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{j}{\sqrt{2}} \end{bmatrix}$, compute the projection of $\vec{v}_1$ onto $\vec{v}_2$ using $P_{\vec{v}_2}$ and the projection of $\vec{v}_2$ onto $\vec{v}_1$ using $P_{\vec{v}_1}$. What are the projection coefficients c_1 and c_2 such that $P_{\vec{v}_2}\vec{v}_1 = c_1\vec{v}_2$, and $P_{\vec{v}_1}\vec{v}_2 = c_2\vec{v}_1$? Do these make sense? How are they related to each other?

- (e) We have everything we need to develop a generic inner product for complex vectors. Given $\vec{u}$ a complex vector, we would like the projection of $\vec{v}$ onto $\vec{u}$ to have coefficient $\frac{\langle \vec{v}, \vec{u} \rangle}{\langle \vec{u}, \vec{u} \rangle}$ (so that the projection is $P_{\vec{u}}\vec{v} = \frac{\langle \vec{v}, \vec{u} \rangle}{\langle \vec{u}, \vec{u} \rangle} \vec{u}$). Furthermore, from the previous parts, we know that the projection coefficient of the projection of $\vec{v}$ onto $\vec{u}$ is the conjugate of the projection coefficient of the projection of $\vec{u}$ onto $\vec{v}$, so we would like $\langle \vec{v}, \vec{u} \rangle = \overline{\langle \vec{u}, \vec{v} \rangle}$.

Verify that the inner product $\langle \vec{v}, \vec{u} \rangle = \vec{u}^* \vec{v}$ satisfies these properties. (Note the switching of the order of the arguments – $\vec{v}$ is first in the inner product notation but second in the conjugate transpose notation!)

In the case of $\vec{u}, \vec{v}$ being real vectors, then $\langle \vec{v}, \vec{u} \rangle = \vec{u}^T \vec{v}$, showing that this inner product reduces to the inner product of real vectors.

One thing to note is that we could have defined the equally valid inner product $\langle \vec{v}, \vec{u} \rangle = \vec{v}^* \vec{u}$. The choice we make is due to popular convention, but importantly *we are going to stick with our choice* of $\langle \vec{v}, \vec{u} \rangle = \vec{u}^* \vec{v}$. Mixing up the inner product order would mess up whatever calculation we're doing.

Anyways, we now have a proposed notion of projection and inner product. Let's make sure they work together and are self-consistent.

- (f) Recall the standard definitions of *orthogonal* and *orthonormal*. Two vectors $\vec{u}$ and $\vec{v}$ are defined to be *orthogonal* when $\langle \vec{u}, \vec{v} \rangle = 0$. Note that in the complex inner product case, when $\langle \vec{u}, \vec{v} \rangle = 0$, $\langle \vec{v}, \vec{u} \rangle = 0$ as well, since $\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$. Two vectors $\vec{u}$ and $\vec{v}$ are *orthonormal* if they are orthogonal and $\|\vec{u}\| = \|\vec{v}\| = 1$.

Let $\vec{u}$ be a complex vector. Verify the basic Pythagorean property that the residual $\vec{r} = \vec{v} - P_{\vec{u}}\vec{v}$ that remains after projecting $\vec{v}$ onto $\vec{u}$ is orthogonal to $\vec{u}$.

- (g) One of the main reasons we study complex inner products in this class is to be able to handle the DFT, which is coming up in the next section. This involves the manipulation of a particular matrix which has orthonormal columns.

Show that if the columns of a square matrix are orthonormal, then the conjugate transpose of the matrix is its inverse. (*Hint: show for a matrix M with orthonormal columns, that $M^*M = I$.*)

The remainder of the worksheet is several examples of complex projections, each with a different type of projection coefficient.

- (h) Let's consider two vectors that aren't linearly dependent. For the two unit vectors $\vec{v}_1 = \begin{bmatrix} j \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{j}{\sqrt{2}} \end{bmatrix}$, compute the projection of $\vec{v}_1$ onto $\vec{v}_2$ and the projection of $\vec{v}_2$ onto $\vec{v}_1$, using either the projection operator or the inner product. What are the coefficients of the projection? Do these make sense? How are they related to each other?

- (i) Here we have another example with complex vectors where the two vectors are linearly dependent, but the vectors are now linearly dependent by a coefficient that is no longer purely imaginary. Specifically, we have the following two vectors $\vec{v}_1 = \begin{bmatrix} \frac{j}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} \frac{-1+j}{2} \\ \frac{-1-j}{2} \end{bmatrix}$.

Compute the projection of $\vec{v}_1$ onto $\vec{v}_2$ and the projection of $\vec{v}_2$ onto $\vec{v}_1$ using either the projection operator or the inner product. What are the projection coefficients? Do these make sense? How are they related to each other?

- (j) Now let's see how to extend this for non-unit vectors. For two vectors $\vec{v}_1 = \begin{bmatrix} j \\ -1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 1 \\ j \end{bmatrix}$, compute the projection of $\vec{v}_1$ onto $\vec{v}_2$ and the projection of $\vec{v}_2$ onto $\vec{v}_1$. What are the projection coefficients? Do these make sense? How are they related to each other?

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EECS 16B Designing Information Devices and Systems II

Discussion Worksheet

Discussion 14B

Questions

1. DFT

In order to get practice with calculating the Discrete Fourier Transform (DFT), this problem will have you calculate the DFT for a few variations on a cosine signal.

Consider a sampled signal that is a function of discrete time $x[t]$. We can represent it as a vector of discrete samples over time $\vec{x}$, of length N .

$$\vec{x} = \begin{bmatrix} x[0] & \dots & x[N-1] \end{bmatrix}^T \quad (1)$$

Let $\vec{X} = \begin{bmatrix} X[0] & \dots & X[N-1] \end{bmatrix}^T$ be the signal $\vec{x}$ represented in the frequency domain, then

$$\vec{x} = U\vec{X} \quad (2)$$

and the inverse operation is given by

$$\vec{X} = U^{-1}\vec{x} = U^*\vec{x} \quad (3)$$

where the columns of U are the orthonormal DFT basis vectors.

$$U = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & e^{j\frac{2\pi}{N}} & e^{j\frac{2\pi(2)}{N}} & \dots & e^{j\frac{2\pi(N-1)}{N}} \\ 1 & e^{j\frac{2\pi(2)}{N}} & e^{j\frac{2\pi(4)}{N}} & \dots & e^{j\frac{2\pi 2(N-1)}{N}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j\frac{2\pi(N-1)}{N}} & e^{j\frac{2\pi 2(N-1)}{N}} & \dots & e^{j\frac{2\pi(N-1)(N-1)}{N}} \end{bmatrix} \quad (4)$$

$$= \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega_N^1 & \omega_N^2 & \dots & \omega_N^{(N-1)} \\ 1 & \omega_N^2 & \omega_N^{2 \cdot 2} & \dots & \omega_N^{(N-1)2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_N^{N-1} & \omega_N^{2(N-1)} & \dots & \omega_N^{(N-1)(N-1)} \end{bmatrix}, \quad (5)$$

where $\omega_N = e^{j\frac{2\pi}{N}}$ is the N th primitive root of unity.

We sometimes call the components of $\vec{X}$ the *DFT coefficients* of the time-domain signal $\vec{x}$. We can think of the components of $\vec{X}$ as weights that represent $\vec{x}$ in the DFT basis.

- (a) Let's begin by looking at the DFT of $x_1[n] = \cos\left(\frac{2\pi}{5}n\right)$ for $N = 5$ samples $n \in \{0, 1, \dots, 4\}$. Compute the DFT basis matrix U .

- (b) Write out $\vec{x}_1$ in terms of the DFT basis vectors.

- (c) Find the DFT coefficients $X_1[k]$.

(d) Plot the time domain representation of $x_1[n]$. Plot the magnitude, $|X_1[k]|$, and plot the phase, $\angle X_1[k]$, for the DFT representation $\vec{X}_1$.

(e) Now let's consider the case where we have a non-zero phase. Let $x_2[n] = \cos\left(\frac{4\pi}{5}n + \pi\right)$. Find the DFT coefficients $\vec{X}_2$ for $\vec{x}_2$.

(f) Plot the time domain representation of $x_2[n]$. Plot the magnitude, $|X_2[k]|$, and plot the phase, $\angle X_2[k]$, for the DFT representation $\vec{X}_2$.

(g) Now let's look at the reverse direction. Given $\vec{X}_3 = \begin{bmatrix} 2 & e^{-j\frac{\pi}{2}} & 0 & 0 & e^{j\frac{\pi}{2}} \end{bmatrix}^T$, find $x_3[n]$.

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